

45. Management Data Input/Output (MDIO) Interface

45.2 MDIO Interface registers

45.2.1 PMA/PMD registers

Change the first sentence of the first paragraph as follows:

For devices operating at 25 Gb/s or higher speeds, the PMA may be instantiated as multiple sublayers (see [83.1.4](#), [109.1.4](#), ~~and [120.1.4](#)~~, [173.1.4](#), and [176.1.5](#) for how MMD addresses are allocated to multiple PMA sublayers for the respective speeds).

Change Table 45–3 (as amended by IEEE Std 802.3cx-2023, IEEE Std 802.3cz-2023 and IEEE Std 802.3df-2024) as follows (most unchanged rows not shown):

Table 45–3—PMA/PMD registers

Register address	Register name	Subclause
...		
1.72	BASE-AU PMA/PMD extended ability	45.2.1.60a
1.73	800G PMA/PMD ability	45.2.1.60b
1.74	800G PMA/PMD extended ability	45.2.1.60c
1.75	400G PMA/PMD extended ability 2	45.2.1.60d
1.76	1.6T PMA/PMD ability	45.2.1.60e
1.77 through 1.79 1.74 through 1.79	Reserved	
...		
1.1458 through 1.1458 1.1458 through 1.1499	Reserved	
1.1459	ILT max wait timer duration	45.2.1.168a
1.1460 through 1.4475	PMA/PMD per lane PRBS31 seeds used for training	45.2.1.169b
1.1476	PMA/PMD interface training status	45.2.1.169c
1.1477	PMA/PMD interface lane training status	45.2.1.169d
1.1478	Reserved	
1.1479	PMA/PMD interface training control	45.2.1.169f
1.1480	RS-FEC symbol multiplexing PMA status	45.2.1.169g
1.1481 through 1.1499	Reserved	
...		
1.1716 through 1.1799	Reserved	
1.1800	TimeSync FEC /PMA/PMD capability	45.2.1.175
...		

Table 45–3—PMA/PMD registers (*continued*)

Register address	Register name	Subclause
1.1811 and 1.1812	TimeSync PMA/PMD receive path data delay in sub-ns	45.2.1.177
1.1813 through 1.1818	TimeSync FEC transmit path data delay registers	45.2.1.177a
1.1819 through 1.1824	TimeSync FEC receive path data delay registers	45.2.1.177b
1.1825	TimeSync FEC configuration	45.2.1.177c
1.1826 through 1.1899 1.1813 through 1.1899	Reserved	
...		
1.2207 through 1.2209	Reserved IFEC codeword counter	45.2.1.220a
1.2210 through 1.2241 1.2210 through 1.2217	IFEC symbol error counter, lane 0 to 15 3	45.2.1.222
1.2242 through 1.2245 1.2218 through 1.2245	Reserved	
...		
1.2318 through 1.2400 1.2318 through 1.32767	Reserved	
1.2401	Inner FEC PRBS13Q testing	45.2.1.251
1.2402	Inner FEC PRBS31 testing	45.2.1.252
1.2403	Inner FEC SSPRQ testing	45.2.1.253
1.2404	Inner FEC square wave testing	45.2.1.254
1.2405	PRBS31 encoded by Inner FEC	45.2.1.255
1.2406 through 1.2409	Reserved	
1.2410	Inner FEC alignment status	45.2.1.256
1.2411	Inner FEC status 1	45.2.1.257
1.2412 through 1.2413	Inner or ER1 FEC lane 0 corrected codewords counter	45.2.1.258
1.2414 through 1.2415	Inner or ER1 FEC lane 0 uncorrected codewords counter	45.2.1.259
1.2416 through 1.2419	Inner or ER1 FEC lane 0 total bits counter	45.2.1.260
1.2420 through 1.2423	Inner or ER1 FEC lane 0 corrected bits counter	45.2.1.261
1.2424 through 1.2426	Inner FEC lane 0 codeword counter	45.2.1.262
1.2427 through 1.2436	Inner FEC lane 0 error bin counters	45.2.1.263
1.2437 through 1.2611	Inner FEC counter registers for lanes 1 through 7	45.2.1.264
1.2612 through 1.2639	Reserved	
1.2640 through 1.2642	ER1 total CRC32 blocks processed	45.2.1.265
1.2643	ER1 CRC32 error counter	45.2.1.266
1.2644 through 1.2694	Inner FEC test block error bin	45.2.1.267
1.2695 through 1.2699	Reserved	
1.2700 through 1.2731	IFEC lane mapping	45.2.1.268 45.2.1.269
1.2732 through 1.2733	IFEC alignment status	45.2.1.270
1.2734 through 1.2763	IFEC codeword error bin counters	45.2.1.271

Table 45–3—PMA/PMD registers (*continued*)

Register address	Register name	Subclause
<u>1.2764 through 1.2769</u>	<u>Reserved</u>	
<u>1.2770 through 1.2801</u>	<u>IFEC symbol error counter, lane 16 to 31</u>	<u>45.2.1.222</u>
<u>1.2802 through 1.2899</u>	<u>Reserved</u>	
<u>1.2900 through 1.3707</u>	<u>PMA transmit test block error bin counters</u>	<u>45.2.1.272</u>
<u>1.3708 through 1.3715</u>	<u>PMA receive test block error bin counters</u>	<u>45.2.1.272</u>
<u>1.3716 through 1.3999</u>	<u>Reserved</u>	
<u>1.4000 through 1.7999</u>	<u>Registers for the upper AUI component (network side of the AUI)</u>	<u>45.2.1.273</u>
<u>1.8000 through 1.32767</u>	<u>Reserved</u>	
...		

45.2.1.1 PMA/PMD control 1 register (Register 1.0)

Change Table 45–4 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–4—PMA/PMD control 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.0.5:2	Speed selection	5 4 3 2 1 1 x x = Reserved 1 0 1 1 = 1.6 Tb/s Reserved 1 0 1 0 = 800 Gb/s 1 0 0 1 = 400 Gb/s 1 0 0 0 = 200 Gb/s 0 1 1 1 = 5 Gb/s 0 1 1 0 = 2.5 Gb/s 0 1 0 1 = 50 Gb/s 0 1 0 0 = 25 Gb/s 0 0 1 1 = 100 Gb/s 0 0 1 0 = 40 Gb/s 0 0 0 1 = 10PASS-TS/2BASE-TL 0 0 0 0 = 10 Gb/s	R/W
...			

^a R/W = Read/Write, SC = Self-clearing, RO = Read only

45.2.1.1.3 Speed selection (1.0.13, 1.0.6, 1.0.5:2)

Change the last paragraph of 45.2.1.1.3 as follows:

When bits 5 through 2 are set to 0010 the use of a 40G PMA/PMD is selected; when set to 0011 the use of a 100G PMA/PMD is selected; when set to 0100 the use of a 25G PMA/PMD is selected; when set to 0101 the use of a 50G PMA/PMD is selected; when set to 0110 the use of a 2.5G PMA/PMD is selected; when set to 0111 the use of a 5G PMA/PMD is selected; when set to 1000 the use of a 200G PMA/PMD is selected; when set to 1001 the use of a 400G PMA/PMD is selected; when set to 1010 the use of an 800G PMA/PMD is selected; when set to 1011 the use of a 1.6T PMA/PMD is selected. More specific selection is performed using the PMA/PMD control 2 register (register 1.7) (see 45.2.1.6.3).

45.2.1.6 PMA/PMD control 2 register (Register 1.7)

Change Table 45–7 (as amended by IEEE Std 802.3df-2024 and IEEE Std 802.3dk-2026) as follows (unchanged table rows and most unchanged bit description lines not shown):

Table 45–7—PMA/PMD control 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.7.7:0	PMA/PMD type selection	7 6 5 4 3 2 1 0 ... <u>1 0 1 0 0 1 0 0 = 100GBASE-BR10-D PMA/PMD</u> <u>1 0 1 0 0 0 1 1 = 1.6TBASE-DR8-2 PMA/PMD</u> <u>1 0 1 0 0 0 1 0 = 1.6TBASE-DR8 PMA/PMD</u> <u>1 0 1 0 0 0 0 1 = 1.6TBASE-CR8 PMA/PMD</u> <u>1 0 1 0 0 0 0 0 = 1.6TBASE-KR8 PMA/PMD</u> <u>1 0 0 1 1 1 x x = reserved</u> <u>1 0 0 1 1 0 1 1 = 400GBASE-DR2-2 PMA/PMD</u> <u>1 0 0 1 1 0 1 0 = 400GBASE-DR2 PMA/PMD</u> <u>1 0 0 1 1 0 0 1 = 400GBASE-CR2 PMA/PMD</u> <u>1 0 0 1 1 0 0 0 = 400GBASE-KR2 PMA/PMD</u> <u>1 0 0 1 0 1 x x = reserved</u> <u>1 0 0 1 0 0 1 1 = 200GBASE-DR1-2 PMA/PMD</u> <u>1 0 0 1 0 0 1 0 = 200GBASE-DR1 PMA/PMD</u> <u>1 0 0 1 0 0 0 1 = 200GBASE-CR1 PMA/PMD</u> <u>1 0 0 1 0 0 0 0 = 200GBASE-KR1 PMA/PMD</u> <u>1 0 0 0 1 1 1 1 = 800GBASE-ER1 PMA/PMD</u> <u>1 0 0 0 1 1 1 0 = 800GBASE-ER1-20 PMA/PMD</u> <u>1 0 0 0 1 1 0 1 = 800GBASE-LR1 PMA/PMD</u> <u>1 0 0 0 1 1 0 0 = 800GBASE-LR4 PMA/PMD</u> <u>1 0 0 0 1 0 1 1 = 800GBASE-FR4 PMA/PMD</u> <u>1 0 0 0 1 0 1 0 = 800GBASE-FR4-500 PMA/PMD</u> <u>1 0 0 0 1 0 0 1 = 800GBASE-DR4-2 PMA/PMD</u> <u>1 0 0 0 1 0 0 0 = 800GBASE-DR4 PMA/PMD</u> <u>1 0 0 0 0 1 1 1 = 800GBASE-CR4 PMA/PMD</u> <u>1 0 0 0 0 1 1 0 = 800GBASE-KR4 PMA/PMD</u> <u>1 0 1 0 0 0 x x = reserved</u> <u>1 0 0 1 x x x x = reserved</u> <u>1 0 0 0 1 x x x = reserved</u> <u>1 0 0 0 0 1 1 x = reserved</u> 1 0 0 0 0 1 0 1 = 800GBASE-DR8-2 PMA/PMD ...	R/W

^a R/W = Read/Write, RO = Read only

45.2.1.7 PMA/PMD status 2 register (Register 1.8)

45.2.1.7.4 Transmit fault (1.8.11)

Change Table 45–9 as follows (most unchanged rows not shown):

Table 45–9—Transmit fault description location

PMA/PMD	Description location
...	
400GBASE-FR4, 400GBASE-LR4-6	151.5.10
<u>200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, 1.6TBASE-KR8</u>	<u>178.8.8</u>
<u>200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8</u>	<u>179.8.8</u>
<u>200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, 1.6TBASE-DR8</u>	<u>180.5.10</u>
<u>800GBASE-FR4-500</u>	<u>181.5.9</u>
<u>200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, 1.6TBASE-DR8-2</u>	<u>182.5.10</u>
<u>800GBASE-FR4, 800GBASE-LR4</u>	<u>183.5.10</u>
<u>800GBASE-LR1</u>	<u>185.5.9</u>
<u>800GBASE-ER1, 800GBASE-ER1-20</u>	<u>187.5.9</u>

45.2.1.7.5 Receive fault (1.8.10)

Change Table 45–10 as follows (most unchanged rows not shown):

Table 45–10—Receive fault description location

PMA/PMD	Description location
...	
400GBASE-FR4, 400GBASE-LR4-6	151.5.11
200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, 1.6TBASE-KR8	178.8.8
200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, 1.6TBASE-CR8	179.8.8
200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, 1.6TBASE-DR8	180.5.11
800GBASE-FR4-500	181.5.10
200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, 1.6TBASE-DR8-2	182.5.11
800GBASE-FR4, 800GBASE-LR4	183.5.11
800GBASE-LR1	185.5.10
800GBASE-ER1, 800GBASE-ER1-20	187.5.10

45.2.1.8 PMD transmit disable register (Register 1.9)

Change Table 45–12 as follows (most unchanged rows not shown):

Table 45–12—Transmit disable description location

PMA/PMD	Description location
...	
400GBASE-FR4 and 400GBASE-LR4-6	151.5.7
<u>200GBASE-KR1, 400GBASE-KR2, 800GBASE-KR4, and 1.6TBASE-KR8</u>	178.8.6
<u>200GBASE-CR1, 400GBASE-CR2, 800GBASE-CR4, and 1.6TBASE-CR8</u>	179.8.6
<u>200GBASE-DR1, 400GBASE-DR2, 800GBASE-DR4, and 1.6TBASE-DR8</u>	180.5.7
<u>800GBASE-FR4-500</u>	181.5.7
<u>200GBASE-DR1-2, 400GBASE-DR2-2, 800GBASE-DR4-2, and 1.6TBASE-DR8-2</u>	182.5.7
<u>800GBASE-FR4 and 800GBASE-LR4</u>	183.5.7
<u>800GBASE-LR1</u>	185.5.7
<u>800GBASE-ER1 and 800GBASE-ER1-20</u>	187.5.7

45.2.1.10 PMA/PMD extended ability register (Register 1.11)

Change Table 45–14 as follows (most unchanged rows not shown):

Table 45–14—PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
1.11.13	200G/400G extended abilities	1 = PMA/PMD has 200G/400G extended abilities listed in register 1.23 (<u>200G</u>) or registers 1.24 and 1.75 (<u>400G</u>) 0 = PMA/PMD does not have 200G/400G extended abilities	RO
...			

^a RO = Read only.

Change 45.2.1.10.3 as follows:

45.2.1.10.3 200G/400G extended abilities (1.11.13)

When read as a one, bit 1.11.13 indicates that the PMA/PMD has 200G extended abilities listed in register 1.23 or 400G extended abilities listed in registers 1.24 and 1.75. When read as a zero, bit 1.11.13 indicates that the PMA/PMD does not have 200G or 400G extended abilities.

45.2.1.21 200G PMA/PMD extended ability register (Register 1.23)

Change Table 45–24 (as amended by IEEE Std 802.3ck-2022) as follows (most unchanged rows not shown):

Table 45–24—200G PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.23.15	200G PMA remote loopback ability	1 = 200G PMA has the ability to perform a remote loopback function 0 = 200G PMA does not have the ability to perform a remote loopback function	RO
1.23.14:11	Reserved	Value always 0	RO
<u>1.23.14</u>	<u>200GBASE-DR1-2 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-DR1-2</u> <u>0 = PMA/PMD is not able to perform 200GBASE-DR1-2</u>	<u>RO</u>
<u>1.23.13</u>	<u>200GBASE-DR1 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-DR1</u> <u>0 = PMA/PMD is not able to perform 200GBASE-DR1</u>	<u>RO</u>
<u>1.23.12</u>	<u>200GBASE-CR1 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-CR1</u> <u>0 = PMA/PMD is not able to perform 200GBASE-CR1</u>	<u>RO</u>
<u>1.23.11</u>	<u>200GBASE-KR1 ability</u>	<u>1 = PMA/PMD is able to perform 200GBASE-KR1</u> <u>0 = PMA/PMD is not able to perform 200GBASE-KR1</u>	<u>RO</u>
1.23.10	200GBASE-SR2 ability	1 = PMA/PMD is able to perform 200GBASE-SR2 0 = PMA/PMD is not able to perform 200GBASE-SR2	RO
...			

^a RO = Read only.

Insert 45.2.1.21.1a to 45.2.1.21.1d after 45.2.1.21.1 as follows:

45.2.1.21.1a 200GBASE-DR1-2 ability (1.23.14)

When read as a one, bit 1.23.14 indicates that the PMA/PMD is able to operate as a 200GBASE-DR1-2 PMA/PMD type. When read as a zero, bit 1.23.14 indicates that the PMA/PMD is not able to operate as a 200GBASE-DR1-2 PMA/PMD type.

45.2.1.21.1b 200GBASE-DR1 ability (1.23.13)

When read as a one, bit 1.23.13 indicates that the PMA/PMD is able to operate as a 200GBASE-DR1 PMA/PMD type. When read as a zero, bit 1.23.13 indicates that the PMA/PMD is not able to operate as a 200GBASE-DR1 PMA/PMD type.

45.2.1.21.1c 200GBASE-CR1 ability (1.23.12)

When read as a one, bit 1.23.12 indicates that the PMA/PMD is able to operate as a 200GBASE-CR1 PMA/PMD type. When read as a zero, bit 1.23.12 indicates that the PMA/PMD is not able to operate as a 200GBASE-CR1 PMA/PMD type.

45.2.1.21.1d 200GBASE-KR1 ability (1.23.11)

When read as a one, bit 1.23.11 indicates that the PMA/PMD is able to operate as a 200GBASE-KR1 PMA/PMD type. When read as a zero, bit 1.23.11 indicates that the PMA/PMD is not able to operate as a 200GBASE-KR1 PMA/PMD type.

45.2.1.23 PMA/PMD extended ability 2 (Register 1.25)

Change Table 45–26 (as amended by IEEE Std 802.3df-2024) as follows:

Table 45–26—PMA/PMD extended ability 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.25.15:3 1.25.15:2	Reserved	Value always 0	RO
<u>1.25.2</u>	<u>1.6T capable</u>	<u>1 = PMA/PMD is capable of operating at 1.6 Tb/s and has the abilities listed in register 1.76</u> <u>0 = PMA/PMD is not capable of operating at 1.6 Tb/s</u>	<u>RO</u>
1.25.1	800G capable	1 = PMA/PMD is capable of operating at 800 Gb/s and has the abilities listed in registers 1.73 and 1.74 0 = PMA/PMD is not capable of operating at 800 Gb/s	RO
1.25.0	50G extended abilities	1 = PMA/PMD has 50G extended abilities listed in register 1.20 0 = PMA/PMD does not have 50G extended abilities	RO

^a RO = Read only

Insert 45.2.1.23.aa before 45.2.1.23.a (as inserted by IEEE Std 802.3df-2024) as follows:

45.2.1.23.aa 1.6T capable (1.25.2)

When read as a one, bit 1.25.2 indicates that the PMA/PMD is capable of operating at 1.6 Tb/s and has the abilities listed in register 1.76. When read as a zero, bit 1.25.2 indicates that the PMA/PMD is not capable of operating at 1.6 Tb/s.

45.2.1.60b 800G PMA/PMD ability (Register 1.73)

Change Table 45–58b (as inserted by IEEE Std 802.3df-2024) as follows:

Table 45–58b—800G PMA/PMD ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.73.15	800G PMA remote loopback ability	1 = 800G PMA is able to perform a remote loopback function 0 = 800G PMA is not able to perform a remote loopback function	RO
1.73.14 1.73.14:6	Reserved	Value always 0	RO
<u>1.73.13</u>	<u>800GBASE-LR1 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-LR1</u> <u>0 = PMA/PMD is not able to perform 800GBASE-LR1</u>	<u>RO</u>
<u>1.73.12</u>	<u>800GBASE-LR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-LR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-LR4</u>	<u>RO</u>
<u>1.73.11</u>	<u>800GBASE-FR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-FR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-FR4</u>	<u>RO</u>
<u>1.73.10</u>	<u>800GBASE-DR4-2 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-DR4-2</u> <u>0 = PMA/PMD is not able to perform 800GBASE-DR4-2</u>	<u>RO</u>
<u>1.73.9</u>	<u>800GBASE-FR4-500 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-FR4-500</u> <u>0 = PMA/PMD is not able to perform 800GBASE-FR4-500</u>	<u>RO</u>
<u>1.73.8</u>	<u>800GBASE-DR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-DR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-DR4</u>	<u>RO</u>
<u>1.73.7</u>	<u>800GBASE-CR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-CR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-CR4</u>	<u>RO</u>
<u>1.73.6</u>	<u>800GBASE-KR4 ability</u>	<u>1 = PMA/PMD is able to perform 800GBASE-KR4</u> <u>0 = PMA/PMD is not able to perform 800GBASE-KR4</u>	<u>RO</u>
1.73.5	800GBASE-DR8-2 ability	1 = PMA/PMD is able to perform 800GBASE-DR8-2 0 = PMA/PMD is not able to perform 800GBASE-DR8-2	RO
1.73.4	800GBASE-DR8 ability	1 = PMA/PMD is able to perform 800GBASE-DR8 0 = PMA/PMD is not able to perform 800GBASE-DR8	RO
1.73.3	800GBASE-SR8 ability	1 = PMA/PMD is able to perform 800GBASE-SR8 0 = PMA/PMD is not able to perform 800GBASE-SR8	RO
1.73.2	800GBASE-VR8 ability	1 = PMA/PMD is able to perform 800GBASE-VR8 0 = PMA/PMD is not able to perform 800GBASE-VR8	RO
1.73.1	800GBASE-CR8 ability	1 = PMA/PMD is able to perform 800GBASE-CR8 0 = PMA/PMD is not able to perform 800GBASE-CR8	RO
1.73.0	800GBASE-KR8 ability	1 = PMA/PMD is able to perform 800GBASE-KR8 0 = PMA/PMD is not able to perform 800GBASE-KR8	RO

^a RO = Read only

Insert 45.2.1.60b.1a to 45.2.1.60b.1i after 45.2.1.60b.1 as follows:

45.2.1.60b.1a 800GBASE-LR1 ability (1.73.13)

When read as a one, bit 1.73.13 indicates that the PMA/PMD is able to operate as an 800GBASE-LR1 PMA/PMD type. When read as a zero, bit 1.73.13 indicates that the PMA/PMD is not able to operate as an 800GBASE-LR1 PMA/PMD type.

45.2.1.60b.1b 800GBASE-LR4 ability (1.73.12)

When read as a one, bit 1.73.12 indicates that the PMA/PMD is able to operate as an 800GBASE-LR4 PMA/PMD type. When read as a zero, bit 1.73.12 indicates that the PMA/PMD is not able to operate as an 800GBASE-LR4 PMA/PMD type.

45.2.1.60b.1c 800GBASE-FR4 ability (1.73.11)

When read as a one, bit 1.73.11 indicates that the PMA/PMD is able to operate as an 800GBASE-FR4 PMA/PMD type. When read as a zero, bit 1.73.11 indicates that the PMA/PMD is not able to operate as an 800GBASE-FR4 PMA/PMD type.

45.2.1.60b.1d 800GBASE-DR4-2 ability (1.73.10)

When read as a one, bit 1.73.10 indicates that the PMA/PMD is able to operate as an 800GBASE-DR4-2 PMA/PMD type. When read as a zero, bit 1.73.10 indicates that the PMA/PMD is not able to operate as an 800GBASE-DR4-2 PMA/PMD type.

45.2.1.60b.1e 800GBASE-FR4-500 ability (1.73.9)

When read as a one, bit 1.73.9 indicates that the PMA/PMD is able to operate as an 800GBASE-FR4-500 PMA/PMD type. When read as a zero, bit 1.73.9 indicates that the PMA/PMD is not able to operate as an 800GBASE-FR4-500 PMA/PMD type.

45.2.1.60b.1f 800GBASE-DR4 ability (1.73.8)

When read as a one, bit 1.73.8 indicates that the PMA/PMD is able to operate as an 800GBASE-DR4 PMA/PMD type. When read as a zero, bit 1.73.8 indicates that the PMA/PMD is not able to operate as an 800GBASE-DR4 PMA/PMD type.

45.2.1.60b.1g 800GBASE-CR4 ability (1.73.7)

When read as a one, bit 1.73.7 indicates that the PMA/PMD is able to operate as an 800GBASE-CR4 PMA/PMD type. When read as a zero, bit 1.73.7 indicates that the PMA/PMD is not able to operate as an 800GBASE-CR4 PMA/PMD type.

45.2.1.60b.1h 800GBASE-KR4 ability (1.73.6)

When read as a one, bit 1.73.6 indicates that the PMA/PMD is able to operate as an 800GBASE-KR4 PMA/PMD type. When read as a zero, bit 1.73.6 indicates that the PMA/PMD is not able to operate as an 800GBASE-KR4 PMA/PMD type.

Insert 45.2.1.60c, 45.2.1.60d, and 45.2.1.60e after 45.2.1.60b (inserted by IEEE Std 802.3df-2024) as follows:

45.2.1.60c 800G PMA/PMD extended ability register (Register 1.74)

The assignment of bits in the 800G PMA/PMD extended ability register is shown in Table 45–58c.

Table 45–58c—800G PMA/PMD extended ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.74.15:2	Reserved	Value always 0	RO
1.74.1	800GBASE-ER1 ability	1 = PMA/PMD is able to perform 800GBASE-ER1 0 = PMA/PMD is not able to perform 800GBASE-ER1	RO
1.74.0	800GBASE-ER1-20 ability	1 = PMA/PMD is able to perform 800GBASE-ER1-20 0 = PMA/PMD is not able to perform 800GBASE-ER1-20	RO

^a RO = Read only.

45.2.1.60c.1 800GBASE-ER1 ability (1.74.1)

When read as a one, bit 1.74.1 indicates that the PMA/PMD is able to operate as a 800GBASE-ER1 PMA/PMD type. When read as a zero, bit 1.74.1 indicates that the PMA/PMD is not able to operate as a 800GBASE-ER1 PMA/PMD type.

45.2.1.60c.2 800GBASE-ER1-20 ability (1.74.0)

When read as a one, bit 1.74.0 indicates that the PMA/PMD is able to operate as a 800GBASE-ER1-20 PMA/PMD type. When read as a zero, bit 1.74.0 indicates that the PMA/PMD is not able to operate as a 800GBASE-ER1-20 PMA/PMD type.

45.2.1.60d 400G PMA/PMD extended ability 2 register (Register 1.75)

The assignment of bits in the 400G PMA/PMD extended ability 2 register is shown in Table 45–58d.

Table 45–58d—400G PMA/PMD extended ability 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.75.15:4	Reserved	Value always 0	RO
1.75.3	400GBASE-DR2-2 ability	1 = PMA/PMD is able to perform 400GBASE-DR2-2 0 = PMA/PMD is not able to perform 400GBASE-DR2-2	RO
1.75.2	400GBASE-DR2 ability	1 = PMA/PMD is able to perform 400GBASE-DR2 0 = PMA/PMD is not able to perform 400GBASE-DR2	RO
1.75.1	400GBASE-CR2 ability	1 = PMA/PMD is able to perform 400GBASE-CR2 0 = PMA/PMD is not able to perform 400GBASE-CR2	RO
1.75.0	400GBASE-KR2 ability	1 = PMA/PMD is able to perform 400GBASE-KR2 0 = PMA/PMD is not able to perform 400GBASE-KR2	RO

^a RO = Read only.

45.2.1.60d.1 400GBASE-DR2-2 ability (1.75.3)

When read as a one, bit 1.75.3 indicates that the PMA/PMD is able to operate as a 400GBASE-DR2-2 PMA/PMD type. When read as a zero, bit 1.75.3 indicates that the PMA/PMD is not able to operate as a 400GBASE-DR2-2 PMA/PMD type.

45.2.1.60d.2 400GBASE-DR2 ability (1.75.2)

When read as a one, bit 1.75.2 indicates that the PMA/PMD is able to operate as a 400GBASE-DR2 PMA/PMD type. When read as a zero, bit 1.75.2 indicates that the PMA/PMD is not able to operate as a 400GBASE-DR2 PMA/PMD type.

45.2.1.60d.3 400GBASE-CR2 ability (1.75.1)

When read as a one, bit 1.75.1 indicates that the PMA/PMD is able to operate as a 400GBASE-CR2 PMA/PMD type. When read as a zero, bit 1.75.1 indicates that the PMA/PMD is not able to operate as a 400GBASE-CR2 PMA/PMD type.

45.2.1.60d.4 400GBASE-KR2 ability (1.75.0)

When read as a one, bit 1.75.0 indicates that the PMA/PMD is able to operate as a 400GBASE-KR2 PMA/PMD type. When read as a zero, bit 1.75.0 indicates that the PMA/PMD is not able to operate as a 400GBASE-KR2 PMA/PMD type.

45.2.1.60e 1.6T PMA/PMD ability register (Register 1.76)

The assignment of bits in the 1.6T PMA/PMD ability register is shown in Table 45–58e.

Table 45–58e—1.6T PMA/PMD ability register bit definitions

Bit(s)	Name	Description	R/W ^a
1.76.15	1.6T PMA remote loopback ability	1 = 1.6T PMA has the ability to perform a remote loopback function 0 = 1.6T PMA does not have the ability to perform a remote loopback function	RO
1.76.14:4	Reserved	Value always 0	RO
1.76.3	1.6TBASE-DR8-2 ability	1 = PMA/PMD is able to perform 1.6TBASE-DR8-2 0 = PMA/PMD is not able to perform 1.6TBASE-DR8-2	RO
1.76.2	1.6TBASE-DR8 ability	1 = PMA/PMD is able to perform 1.6TBASE-DR8 0 = PMA/PMD is not able to perform 1.6TBASE-DR8	RO
1.76.1	1.6TBASE-CR8 ability	1 = PMA/PMD is able to perform 1.6TBASE-CR8 0 = PMA/PMD is not able to perform 1.6TBASE-CR8	RO
1.76.0	1.6TBASE-KR8 ability	1 = PMA/PMD is able to perform 1.6TBASE-KR8 0 = PMA/PMD is not able to perform 1.6TBASE-KR8	RO

^a RO = Read only.

45.2.1.60e.1 1.6T PMA remote loopback ability (1.76.15)

When read as a one, bit 1.76.15 indicates that the 1.6T PMA is able to perform the remote loopback function. When read as a zero, bit 1.76.15 indicates that the 1.6T PMA is not able to perform the remote loopback function. If a PMA is able to perform the remote loopback function, then it is controlled using the PMA remote loopback bit 1.0.1 (see 45.2.1.1.4).

45.2.1.60e.2 1.6TBASE-DR8-2 ability (1.76.3)

When read as a one, bit 1.76.3 indicates that the PMA/PMD is able to operate as a 1.6TBASE-DR8-2 PMA/PMD type. When read as a zero, bit 1.76.3 indicates that the PMA/PMD is not able to operate as a 1.6TBASE-DR8-2 PMA/PMD type.

45.2.1.60e.3 1.6TBASE-DR8 ability (1.76.2)

When read as a one, bit 1.76.2 indicates that the PMA/PMD is able to operate as a 1.6TBASE-DR8 PMA/PMD type. When read as a zero, bit 1.76.2 indicates that the PMA/PMD is not able to operate as a 1.6TBASE-DR8 PMA/PMD type.

45.2.1.60e.4 1.6TBASE-CR8 ability (1.76.1)

When read as a one, bit 1.76.1 indicates that the PMA/PMD is able to operate as a 1.6TBASE-CR8 PMA/PMD type. When read as a zero, bit 1.76.1 indicates that the PMA/PMD is not able to operate as a 1.6TBASE-CR8 PMA/PMD type.

45.2.1.60e.5 1.6TBASE-KR8 ability (1.76.0)

When read as a one, bit 1.76.0 indicates that the PMA/PMD is able to operate as a 1.6TBASE-KR8 PMA/PMD type. When read as a zero, bit 1.76.0 indicates that the PMA/PMD is not able to operate as a 1.6TBASE-KR8 PMA/PMD type.

45.2.1.96 BASE-R PMD status register (Register 1.151)

Change 45.2.1.96.1 through 45.2.1.96.4 as follows:

45.2.1.96.1 Receiver status 0 (1.151.0)

This bit maps to the state variable rx_trained as defined in 72.6.10.3.1, ~~and~~ local_trained in 136.8.11.7.1, ~~and~~ local_rx_ready in 178B.8.3.1.

45.2.1.96.2 Frame lock 0 (1.151.1)

This bit maps to the state variable frame_lock as defined in 72.6.10.3.1, ~~and~~ local_tf_lock in 136.8.11.7.1 ~~and~~ 178B.8.3.1.

45.2.1.96.3 Startup protocol status 0 (1.151.2)

This bit maps to the state variable training as defined in 72.6.10.3.1, ~~and~~ 136.8.11.7.1, ~~and~~ 178B.8.3.1.

45.2.1.96.4 Training failure 0 (1.151.3)

This bit maps to the state variable training_failure as defined in 72.6.10.3.1, ~~and~~ 136.8.11.7.1, ~~and~~ 178B.8.3.1.

45.2.1.139 PMA precoder control Tx output (Register 1.600)

Change Table 45–109 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–109—PMA precoder control Tx output register bit definitions

Bit(s)	Name	Description	R/W ^a
1.600.15:8	Reserved	Value always 0	RO
<u>1.600.15</u>	<u>Lane 15 Tx output precoder enable</u>	<u>1 = Lane 15 Tx output precoder enabled</u> <u>0 = Lane 15 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.14</u>	<u>Lane 14 Tx output precoder enable</u>	<u>1 = Lane 14 Tx output precoder enabled</u> <u>0 = Lane 14 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.13</u>	<u>Lane 13 Tx output precoder enable</u>	<u>1 = Lane 13 Tx output precoder enabled</u> <u>0 = Lane 13 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.12</u>	<u>Lane 12 Tx output precoder enable</u>	<u>1 = Lane 12 Tx output precoder enabled</u> <u>0 = Lane 12 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.11</u>	<u>Lane 11 Tx output precoder enable</u>	<u>1 = Lane 11 Tx output precoder enabled</u> <u>0 = Lane 11 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.10</u>	<u>Lane 10 Tx output precoder enable</u>	<u>1 = Lane 10 Tx output precoder enabled</u> <u>0 = Lane 10 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.9</u>	<u>Lane 9 Tx output precoder enable</u>	<u>1 = Lane 9 Tx output precoder enabled</u> <u>0 = Lane 9 Tx output precoder disabled</u>	<u>R/W</u>
<u>1.600.8</u>	<u>Lane 8 Tx output precoder enable</u>	<u>1 = Lane 8 Tx output precoder enabled</u> <u>0 = Lane 8 Tx output precoder disabled</u>	<u>R/W</u>

^aR/W = Read/Write, RO = Read only

Replace 45.2.1.139.a to 45.2.1.139.4 with 45.2.1.139.1 as follows:

45.2.1.139.1 Tx output precoder lane enable

The bits in PMA register 1.600 enable the output precoder on individual lanes in the transmit direction.

45.2.1.140 PMA precoder control Rx input (Register 1.601)

Change Table 45–110 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–110—PMA precoder control Rx input register bit definitions

Bit(s)	Name	Description	R/W ^a
1.601.15:8	Reserved	Value always 0	RO
<u>1.601.15</u>	<u>Lane 15 Rx input precoder enable</u>	<u>1 = Lane 15 Rx input precoder enabled</u> <u>0 = Lane 15 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.14</u>	<u>Lane 14 Rx input precoder enable</u>	<u>1 = Lane 14 Rx input precoder enabled</u> <u>0 = Lane 14 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.13</u>	<u>Lane 13 Rx input precoder enable</u>	<u>1 = Lane 13 Rx input precoder enabled</u> <u>0 = Lane 13 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.12</u>	<u>Lane 12 Rx input precoder enable</u>	<u>1 = Lane 12 Rx input precoder enabled</u> <u>0 = Lane 12 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.11</u>	<u>Lane 11 Rx input precoder enable</u>	<u>1 = Lane 11 Rx input precoder enabled</u> <u>0 = Lane 11 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.10</u>	<u>Lane 10 Rx input precoder enable</u>	<u>1 = Lane 10 Rx input precoder enabled</u> <u>0 = Lane 10 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.9</u>	<u>Lane 9 Rx input precoder enable</u>	<u>1 = Lane 9 Rx input precoder enabled</u> <u>0 = Lane 9 Rx input precoder disabled</u>	<u>R/W</u>
<u>1.601.8</u>	<u>Lane 8 Rx input precoder enable</u>	<u>1 = Lane 8 Rx input precoder enabled</u> <u>0 = Lane 8 Rx input precoder disabled</u>	<u>R/W</u>

^aR/W = Read/Write, RO = Read only

Replace 45.2.1.140.a to 45.2.1.140.4 with 45.2.1.140.1 as follows:

45.2.1.140.1 Rx input precoder lane enable

The bits in PMA register 1.601 enable the input precoder on individual lanes in the receive direction.

45.2.1.141 PMA precoder control Rx output (Register 1.602)

Change Table 45–111 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–111—PMA precoder control Rx output register bit definitions

Bit(s)	Name	Description	R/W ^a
1.602.15:8	Reserved	Value always 0	RO
<u>1.602.15</u>	<u>Lane 15 Rx output precoder enable</u>	<u>1 = Lane 15 Rx output precoder enabled</u> <u>0 = Lane 15 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.14</u>	<u>Lane 14 Rx output precoder enable</u>	<u>1 = Lane 14 Rx output precoder enabled</u> <u>0 = Lane 14 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.13</u>	<u>Lane 13 Rx output precoder enable</u>	<u>1 = Lane 13 Rx output precoder enabled</u> <u>0 = Lane 13 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.12</u>	<u>Lane 12 Rx output precoder enable</u>	<u>1 = Lane 12 Rx output precoder enabled</u> <u>0 = Lane 12 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.11</u>	<u>Lane 11 Rx output precoder enable</u>	<u>1 = Lane 11 Rx output precoder enabled</u> <u>0 = Lane 11 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.10</u>	<u>Lane 10 Rx output precoder enable</u>	<u>1 = Lane 10 Rx output precoder enabled</u> <u>0 = Lane 10 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.9</u>	<u>Lane 9 Rx output precoder enable</u>	<u>1 = Lane 9 Rx output precoder enabled</u> <u>0 = Lane 9 Rx output precoder disabled</u>	<u>R/W</u>
<u>1.602.8</u>	<u>Lane 8 Rx output precoder enable</u>	<u>1 = Lane 8 Rx output precoder enabled</u> <u>0 = Lane 8 Rx output precoder disabled</u>	<u>R/W</u>

^aR/W = Read/Write, RO = Read only

Replace 45.2.1.141.a to 45.2.1.141.4 with 45.2.1.141.1 as follows:

45.2.1.141.1 Rx output precoder lane enable

The bits in PMA register 1.602 enable the output precoder on individual lanes in the receive direction.

45.2.1.142 PMA precoder control Tx input (Register 1.603)

Change Table 45–112 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–112—PMA precoder control Tx input register bit definitions

Bit(s)	Name	Description	R/W ^a
1.603.15:8	Reserved	Value always 0	RO
<u>1.603.15</u>	<u>Lane 15 Tx input precoder enable</u>	<u>1 = Lane 15 Tx input precoder enabled</u> <u>0 = Lane 15 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.14</u>	<u>Lane 14 Tx input precoder enable</u>	<u>1 = Lane 14 Tx input precoder enabled</u> <u>0 = Lane 14 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.13</u>	<u>Lane 13 Tx input precoder enable</u>	<u>1 = Lane 13 Tx input precoder enabled</u> <u>0 = Lane 13 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.12</u>	<u>Lane 12 Tx input precoder enable</u>	<u>1 = Lane 12 Tx input precoder enabled</u> <u>0 = Lane 12 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.11</u>	<u>Lane 11 Tx input precoder enable</u>	<u>1 = Lane 11 Tx input precoder enabled</u> <u>0 = Lane 11 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.10</u>	<u>Lane 10 Tx input precoder enable</u>	<u>1 = Lane 10 Tx input precoder enabled</u> <u>0 = Lane 10 Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.9</u>	<u>Lane 9 Tx input precoder enable</u>	<u>1 = Lane 9 Tx input precoder enabled</u> <u>0 = Lane 9Tx input precoder disabled</u>	<u>R/W</u>
<u>1.603.8</u>	<u>Lane 8 Tx input precoder enable</u>	<u>1 = Lane 8 Tx input precoder enabled</u> <u>0 = Lane 8 Tx input precoder disabled</u>	<u>R/W</u>

^aR/W = Read/Write, RO = Read only

Replace 45.2.1.142.a to 45.2.1.142.4 with 45.2.1.142.1 as follows:

45.2.1.142.1 Tx input precoder lane enable

The bits in PMA register 1.603 enable the input precoder on individual lanes in the transmit direction.

45.2.1.144 PMA precoder request Rx input status (Register 1.605)

Change Table 45–114 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–114—PMA precoder request Rx input status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.605.15:8	Reserved	Value always 0	RO
<u>1.605.15</u>	<u>Lane 15 Rx input precoder request status</u>	<u>1 = Lane 15 precoder Rx input requested</u> <u>0 = Lane 15 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.14</u>	<u>Lane 14 Rx input precoder request status</u>	<u>1 = Lane 14 precoder Rx input requested</u> <u>0 = Lane 14 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.13</u>	<u>Lane 13 Rx input precoder request status</u>	<u>1 = Lane 13 precoder Rx input requested</u> <u>0 = Lane 13 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.12</u>	<u>Lane 12 Rx input precoder request status</u>	<u>1 = Lane 12 precoder Rx input requested</u> <u>0 = Lane 12 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.11</u>	<u>Lane 11 Rx input precoder request status</u>	<u>1 = Lane 11 precoder Rx input requested</u> <u>0 = Lane 11 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.10</u>	<u>Lane 10 Rx input precoder request status</u>	<u>1 = Lane 10 precoder Rx input requested</u> <u>0 = Lane 10 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.9</u>	<u>Lane 9 Rx input precoder request status</u>	<u>1 = Lane 9 precoder Rx input requested</u> <u>0 = Lane 9 precoder Rx input not requested</u>	<u>RO</u>
<u>1.605.8</u>	<u>Lane 8 Rx input precoder request status</u>	<u>1 = Lane 8 precoder Rx input requested</u> <u>0 = Lane 8 precoder Rx input not requested</u>	<u>RO</u>

^aRO = Read only

Replace 45.2.1.144.a to 45.2.1.144.4 with 45.2.1.144.1 as follows:

45.2.1.144.1 Rx input precoder request status

The bits in PMA register 1.605 indicate the input precoder request status on individual lanes in the receive direction.

45.2.1.145 PMA precoder request Tx input status (Register 1.606)

Change Table 45–115 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–115—PMA precoder request Tx input status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.606.15:8	Reserved	Value always 0	RO
<u>1.606.15</u>	<u>Lane 15 Tx input precoder request status</u>	<u>1 = Lane 15 precoder Tx input requested</u> <u>0 = Lane 15 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.14</u>	<u>Lane 14 Tx input precoder request status</u>	<u>1 = Lane 14 precoder Tx input requested</u> <u>0 = Lane 14 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.13</u>	<u>Lane 13 Tx input precoder request status</u>	<u>1 = Lane 13 precoder Tx input requested</u> <u>0 = Lane 13 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.12</u>	<u>Lane 12 Tx input precoder request status</u>	<u>1 = Lane 12 precoder Tx input requested</u> <u>0 = Lane 12 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.11</u>	<u>Lane 11 Tx input precoder request status</u>	<u>1 = Lane 11 precoder Tx input requested</u> <u>0 = Lane 11 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.10</u>	<u>Lane 10 Tx input precoder request status</u>	<u>1 = Lane 10 precoder Tx input requested</u> <u>0 = Lane 10 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.9</u>	<u>Lane 9 Tx input precoder request status</u>	<u>1 = Lane 9 precoder Tx input requested</u> <u>0 = Lane 9 precoder Tx input not requested</u>	<u>RO</u>
<u>1.606.8</u>	<u>Lane 8 Tx input precoder request status</u>	<u>1 = Lane 8 precoder Tx input requested</u> <u>0 = Lane 8 precoder Tx input not requested</u>	<u>RO</u>

^aRO = Read only

Replace 45.2.1.145.a to 45.2.1.145.4 with 45.2.1.145.1 as follows:

45.2.1.145.1 Tx input precoder request status

The bits in PMA register 1.606 indicate the input precoder request status on individual lanes in the transmit direction.

45.2.1.161 BASE-R PAM4 PMD training LP control, lane 0 through lane 7 registers (Register 1.1120 through 1.1127)

Change Table 45–129 (as amended by IEEE Std 802.3ck-2022) as follows:

**Table 45–129—BASE-R PAM4 PMD training LP control, lane 0 register
bit definitions**

Bit(s)	Name	Description	R/W ^a
1.1120.15:14	Reserved	Value always 0	RO
1.1120.13:11	Initial condition request ^b	13 12 11 1 1 1 = Reserved 1 0 1 = Reserved Preset 6 0 1 1 = Preset 5 0 0 1 = Preset 4 1 1 0 = Preset 3 1 0 0 = Preset 2 0 1 0 = Preset 1 0 0 0 = Individual coefficient control	R/W
1.1120.10	Reserved Not ready to send	Value always 0 1 = Not ready to send 0 = Switch to data when training is completed	RO R/W
1.1120.9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	R/W
1.1120.7:5	Reserved	Value always 0	RO
1.1120.6:5	Training pattern request	6 5 1 1 = Free-running PRBS31 1 0 = Reserved 0 1 = Free-running PRBS13 0 0 = Synchronous PRBS13	R/W
1.1120.4:2	Coefficient select ^c	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	R/W
1.1120.1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold	R/W

^a R/W = Read/Write, RO = Read only.

^b Support for a given preset is PHY dependent.

^c Support for a given coefficient is PHY dependent.

45.2.1.163 BASE-R PAM4 PMD training LP status, lane 0 through lane 7 registers (Register 1.1220 through 1.1227)

Change Table 45–130 (as amended by IEEE Std 802.3ck-2022) as follows:

Table 45–130—BASE-R PAM4 PMD training LP status, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1220.15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue	RO
1.1220.14:12	Reserved	Value always 0	RO
<u>1.1220.13:12</u>	<u>Training pattern status</u>	<u>13 12</u> <u>1 1</u> = Free-running PRBS31 <u>1 0</u> = Reserved <u>0 1</u> = Free-running PRBS13 <u>0 0</u> = Synchronous PRBS13	<u>RO</u>
1.1220.11:10	Modulation and precoding status	11 10 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	RO
1.1220.9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified	RO
1.1220.8	Initial condition status	1 = Updated 0 = Not updated	RO
1.1220.7	Parity	Even parity bit	RO
1.1220.6	Reserved	Value always 0	RO
1.1220.5:3	Coefficient select echo ^b	5 4 3 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	RO
1.1220.2:0	Coefficient status	2 1 0 1 1 1 = Reserved 1 1 0 = Coefficient at limit and equalization limit 1 0 1 = Reserved 1 0 0 = Equalization limit 0 1 1 = Coefficient not supported 0 1 0 = Coefficient at limit 0 0 1 = Updated 0 0 0 = Not updated	RO

^a RO = Read only.

^b Support for a given coefficient is PHY dependent.

45.2.1.165 BASE-R PAM4 PMD training LD control, lane 0 through lane 7 registers (Register 1.1320 through 1.1327)

Change Table 45–131 (as amended by IEEE Std 802.3ck-2022) as follows:

Table 45–131—BASE-R PAM4 PMD training LD control, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1320.15:14	Reserved	Value always 0	RO
1.1320.13:11	Initial condition request ^b	13 12 11 1 1 1 = Reserved 1 0 1 = Reserved Preset 6 0 1 1 = Preset 5 0 0 1 = Preset 4 1 1 0 = Preset 3 1 0 0 = Preset 2 0 1 0 = Preset 1 0 0 0 = Individual coefficient control	R/W
1.1320.10	Reserved Not ready to send	Value always 0 1 = Not ready to send 0 = Switch to data when training is completed	RO R/W
1.1320.9:8	Modulation and precoding request	9 8 1 1 = PAM4 with precoding 1 0 = PAM4 0 1 = Reserved 0 0 = PAM2	R/W
1.1320.7:5	Reserved	Value always 0	RO
<u>1.1320.6:5</u>	<u>Training pattern request</u>	<u>6 5</u> <u>1 1</u> = Free-running PRBS31 <u>1 0</u> = Reserved <u>0 1</u> = Free-running PRBS13 <u>0 0</u> = Synchronous PRBS13	<u>R/W</u>
1.1320.4:2	Coefficient select ^c	4 3 2 1 0 0 = Reserved 1 0 1 = $c(-3)$ 1 1 0 = $c(-2)$ 1 1 1 = $c(-1)$ 0 0 0 = $c(0)$ 0 0 1 = $c(1)$ 0 1 x = Reserved	R/W
1.1320.1:0	Coefficient request	1 0 1 1 = No equalization 1 0 = Decrement 0 1 = Increment 0 0 = Hold	R/W

^a R/W = Read/Write, RO = Read only.

^b Support for a given preset is PHY dependent.

^c Support for a given coefficient is PHY dependent.

45.2.1.167 BASE-R PAM4 PMD training LD status, lane 0 through lane 7 registers (Register 1.1420 through 1.1427)

Change Table 45–132 (as amended by IEEE Std 802.3ck-2022) as follows:

Table 45–132—BASE-R PAM4 PMD training LD status, lane 0 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1420.15	Receiver ready	1 = Training is complete and the receiver is ready for data 0 = Request for training to continue	RO
1.1420.14:12	Reserved	Value always 0	RO
<u>1.1420.13:12</u>	<u>Training pattern status</u>	$\begin{matrix} 13 & 12 \\ 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{matrix}$ $\begin{matrix} = \text{Free-running PRBS31} \\ = \text{Reserved} \\ = \text{Free-running PRBS13} \\ = \text{Synchronous PRBS13} \end{matrix}$	<u>RO</u>
1.1420.11:10	Modulation and precoding status	$\begin{matrix} 11 & 10 \\ 1 & 1 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{matrix}$ $\begin{matrix} = \text{PAM4 with precoding} \\ = \text{PAM4} \\ = \text{Reserved} \\ = \text{PAM2} \end{matrix}$	RO
1.1420.9	Receiver frame lock	1 = Frame boundaries identified 0 = Frame boundaries not identified	RO
1.1420.8	Initial condition status	1 = Updated 0 = Not updated	RO
1.1420.7	Parity	Even parity bit	RO
1.1420.6	Reserved	Value always 0	RO
1.1420.5:3	Coefficient select echo ^b	$\begin{matrix} 5 & 4 & 3 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & x \end{matrix}$ $\begin{matrix} = \text{Reserved} \\ = c(-3) \\ = c(-2) \\ = c(-1) \\ = c(0) \\ = c(1) \\ = \text{Reserved} \end{matrix}$	RO
1.1420.2:0	Coefficient status	$\begin{matrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{matrix}$ $\begin{matrix} = \text{Reserved} \\ = \text{Coefficient at limit and equalization limit} \\ = \text{Reserved} \\ = \text{Equalization limit} \\ = \text{Coefficient not supported} \\ = \text{Coefficient at limit} \\ = \text{Updated} \\ = \text{Not updated} \end{matrix}$	RO

^a RO = Read only.

^b Support for a given coefficient is PHY dependent.

Insert 45.2.1.168a through 45.2.1.168f after 45.2.1.168 as follows:

45.2.1.168a ILT max wait timer duration register (Register 1.1459)

Table 45–133a shows the assignment of bits for the ILT max wait time duration register.

Table 45–133a—ILT max wait timer duration register bit definitions

Bit(s)	Name	Description	R/W ^a
1.1459.15:0	ILT max wait timer duration	ILT_max_wait_timer_duration[15:0]	R/W

^a R/W = Read /Write

45.2.1.169b PMA/PMD PRBS31 seed value lanes 0 through 7 (Register 1.1460 through 1.1475)

The assignment of bits in the PMA/PMD PRBS31 seed value lane 0 register is shown in Table 45–133b. The assignment of bits in the PMA/PMD PRBS31 seed value lanes 1 through 7 registers are defined similarly to lane 0. Registers 1.1460 and 1.1461 define the seed for PMA/PMD lane 0; registers 1.1462 and 1.1463 define the seed for lane 1; etc.

Table 45–133b—PRBS31 seed value lane 0 bit definitions

Bit(s)	Name	Description	R/W ^a
1.1460.15:0	PRBS31 seed lower	Bits 15:0 of the PRBS31 binary seed used for lane 0 training pattern	R/W
1.1461.15	Reserved	Value always 0	RO
1.1461.14:0	PRBS31 seed upper	Bits 30:16 of the PRBS31 binary seed used for lane 0 training pattern	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.169c PMA/PMD interface training status (Register 1.1476)

The assignment of bits in the PMA/PMD interface training status register is shown in Table 45–133c.

Table 45–133c—PMA/PMD interface training status bit definitions

Bit(s)	Name	Description	R/W ^a
1.1476.15	Reserved	Value always 0	RO
1.1476.14:13	Ready to send	14 13 1 1 = rts_status is OK 1 0 = rts_status is READY 0 1 = rts_status is IN_PROGRESS 0 0 = rts_status is FAIL	RO
1.1476.12	Reserved	Value always 0	RO
1.1476.11:10	PMA/PMD interface training status	11 10 1 1 = training_status is OK 1 0 = training_status is READY 0 1 = training_status is IN_PROGRESS 0 0 = training_status is FAIL	RO
1.1476.9	Reserved	Value always 0	RO
1.1476.8	isl ready	1 = local_rx_ready and remote_rx_ready are true for all lanes of the interface 0 = local_rx_ready or remote_rx_ready is false on any of the lanes of the interface	RO
1.1476.7	Polarity inversion lane 7	1 = polarity inversion detected on lane 7 0 = polarity inversion not detected on lane 7	RO
1.1476.6	Polarity inversion lane 6	1 = polarity inversion detected on lane 6 0 = polarity inversion not detected on lane 6	RO
1.1476.5	Polarity inversion lane 5	1 = polarity inversion detected on lane 5 0 = polarity inversion not detected on lane 5	RO
1.1476.4	Polarity inversion lane 4	1 = polarity inversion detected on lane 4 0 = polarity inversion not detected on lane 4	RO
1.1476.3	Polarity inversion lane 3	1 = polarity inversion detected on lane 3 0 = polarity inversion not detected on lane 3	RO
1.1476.2	Polarity inversion lane 2	1 = polarity inversion detected on lane 2 0 = polarity inversion not detected on lane 2	RO
1.1476.1	Polarity inversion lane 1	1 = polarity inversion detected on lane 1 0 = polarity inversion not detected on lane 1	RO
1.1476.0	Polarity inversion lane 0	1 = polarity inversion detected on lane 0 0 = polarity inversion not detected on lane 0	RO

^a R/W = Read/Write, RO = Read only.

45.2.1.169d PMA/PMD interface lane training status 1 (Register 1.1477)

The assignment of bits in the PMA/PMD interface lane training status 1 register is shown in Table 45–133d.

Table 45–133d—PMA/PMD interface lane training status 1 bit definitions

Bit(s)	Name	Description	R/W ^a
1.1477.15:14	Training status lane 7	15 14 1 1 = lane_training_status is OK 1 0 = lane_training_status is TRAINED 0 1 = lane_training_status is IN_PROGRESS 0 0 = lane_training_status is FAIL	RO
1.1477.13:12	Training status lane 6	13 12 1 1 = lane_training_status is OK 1 0 = lane_training_status is TRAINED 0 1 = lane_training_status is IN_PROGRESS 0 0 = lane_training_status is FAIL	RO
1.1477.11:10	Training status lane 5	11 10 1 1 = lane_training_status is OK 1 0 = lane_training_status is TRAINED 0 1 = lane_training_status is IN_PROGRESS 0 0 = lane_training_status is FAIL	RO
1.1477.9:8	Training status lane 4	9 8 1 1 = lane_training_status is OK 1 0 = lane_training_status is TRAINED 0 1 = lane_training_status is IN_PROGRESS 0 0 = lane_training_status is FAIL	RO
1.1477.7:6	Training status lane 3	7 6 1 1 = lane_training_status is OK 1 0 = lane_training_status is TRAINED 0 1 = lane_training_status is IN_PROGRESS 0 0 = lane_training_status is FAIL	RO
1.1477.5:4	Training status lane 2	5 4 1 1 = lane_training_status is OK 1 0 = lane_training_status is TRAINED 0 1 = lane_training_status is IN_PROGRESS 0 0 = lane_training_status is FAIL	RO
1.1477.3:2	Training status lane 1	3 2 1 1 = lane_training_status is OK 1 0 = lane_training_status is TRAINED 0 1 = lane_training_status is IN_PROGRESS 0 0 = lane_training_status is FAIL	RO
1.1477.1:0	Training status lane 0	1 0 1 1 = lane_training_status is OK 1 0 = lane_training_status is TRAINED 0 1 = lane_training_status is IN_PROGRESS 0 0 = lane_training_status is FAIL	RO

^a R/W = Read/Write, RO = Read only.

45.2.1.169e PMA/PMD interface lane training status 2 (Register 1.1478)

The assignment of bits in the PMA/PMD interface lane training status 2 register is shown in Table 45–133e.

Table 45–133e—PMA/PMD interface lane training status 2 bit definitions

Bit(s)	Name	Description	R/W ^a
1.1478.15:8	Reserved	Value always 0	RO
1.1478.7	Receiver OK lane 7	1 = rx_ok is true 0 = rx_ok is false	RO
1.1478.6	Receiver OK lane 6	1 = rx_ok is true 0 = rx_ok is false	RO
1.1478.5	Receiver OK lane 5	1 = rx_ok is true 0 = rx_ok is false	RO
1.1478.4	Receiver OK lane 4	1 = rx_ok is true 0 = rx_ok is false	RO
1.1478.3	Receiver OK lane 3	1 = rx_ok is true 0 = rx_ok is false	RO
1.1478.2	Receiver OK lane 2	1 = rx_ok is true 0 = rx_ok is false	RO
1.1478.1	Receiver OK lane 1	1 = rx_ok is true 0 = rx_ok is false	RO
1.1478.0	Receiver OK lane 0	1 = rx_ok is true 0 = rx_ok is false	RO

^a R/W = Read/Write, RO = Read only.

45.2.1.169f PMA/PMD interface training control (Register 1.1479)

The assignment of bits in the PMA/PMD interface training control register is shown in Table 45–133f.

Table 45–133f—PMA/PMD interface training control bit definitions

Bit(s)	Name	Description	R/W ^a
1.1479.15	Receiver recovery ability	0 = supports recovery from loss of signal without training restart 1 = requires restart of training for recovery from loss of signal	RO
1.1479.14	Adjacent loss of signal squelch ability	0 = does not support mr_adj_loss_mode = SQUELCH 1 = supports mr_adj_loss_mode = SQUELCH	RO
1.1479.13	Adjacent loss of signal local pattern ability	0 = does not support mr_adj_loss_mode = LOCAL_PATTERN 1 = supports mr_adj_loss_mode = LOCAL_PATTERN	RO
1.1479.12:11	Reserved	Value always 0	RO
1.1479.10:9	Adjacent loss of signal mode	10 9 1 1 = Reserved 1 0 = mr_adj_loss_mode is SQUELCH 0 1 = mr_adj_loss_mode is LOCAL_PATTERN 0 0 = mr_adj_loss_mode is FAIL	R/W
1.1479.8	Uses recovered clock	1 = Use recovered clock for transmission 0 = Use local clock for transmission	R/W
1.1479.7:0	Maximum recovery events	Maximum allowed number of transitions into the RECOVERY state in the training control state diagram (Figure 178B–10a and Figure 178B–10b)	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.169g RS-FEC symbol multiplexing PMA status (Register 1.1480)

The assignment of bits in the RS-FEC symbol multiplexing PMA status register is shown in Table 45–133g.

Table 45–133g—RS-FEC symbol multiplexing PMA status bit definitions

Bit(s)	Name	Description	R/W ^a
1.1480.15:14	Reserved	Value always 0	RO
1.1480.13	align_status_mux	1 = PCS lane synchronization is complete. This bit indicates that all_locked_mux is true and deskew is complete 0 = PCS lane synchronization is not complete.	RO
1.1480.12	all_locked_mux	1 = all_locked as defined in 119.2.6.2.2 is true 0 = all_locked is not true	RO
1.1480.11	deskew_done_mux	1 = deskew is complete 0 = deskew is not complete	RO
1.1480.10	pcs_lanes_identified_mux	1 = all input lanes are locked to a unique alignment marker sequence 0 = not all input lanes are locked to a unique alignment marker sequence	RO
1.1480.9	all_locked_demux	1 = pmal_locked_demux is true for all lanes 0 = pmal_locked_demux is not true for all lanes	RO
1.1480.8	pcs_lanes_identified_demux	1 = all demultiplexed PCS lanes are locked to a unique alignment marker sequence 0 = not all demultiplexed PCS lanes are locked to a unique alignment marker sequence	RO
1.1480.7:0	pmal_locked_demux[7:0]	1 = amps_lock is true for all PCSLs within the indicated input lane in the demultiplexing direction 0 = amps_lock is not true for all PCSLs within the indicated input lane in the demultiplexing direction	RO

^a R/W = Read/Write, RO = Read only.

Change 45.2.1.175 (as amended by IEEE Std 802.3cx-2023) as follows:

45.2.1.175 TimeSync FEC/PMA/PMD capability (Register 1.1800)

The TimeSync FEC/PMA/PMD capability register (see Table 45–139) indicates the capability of the PMA/PMD, Inner FEC, and ER1 FEC to report the transmit path data delay (with ns resolution in registers 1.1801 through 1.1804 and, separately, with sub-ns resolution in registers 1.1809 and 1.1810) and receive

path data delay (with ns resolution in registers 1.1805 through 1.1808 and, separately, with sub-ns resolution in registers 1.1811 and 1.1812). See 90.7.

Table 45–139—TimeSync FEC/PMA/PMD capability

Bit(s)	Name	Description	R/W ^a
<u>1.1800.15:9</u> 1.1800.15:4	Reserved	Value always 0	RO
<u>1.1800.8</u>	<u>FEC alignment marker location</u>	<u>1 = FEC supports alignment marker location</u> <u>0 = FEC does not support alignment marker location</u>	<u>RO</u>
<u>1.1800.7</u>	<u>TimeSync Inner FEC or ER1 FEC transmit path data delay ability, in sub-ns</u>	<u>1 = Inner FEC or ER1 FEC provides information on transmit path data delay with sub-ns resolution in registers 1.1817 and 1.1818</u> <u>0 = Inner FEC or ER1 FEC does not provide information on transmit path data delay with sub-ns resolution</u>	<u>RO</u>
<u>1.1800.6</u>	<u>TimeSync Inner FEC or ER1 FEC transmit path data delay ability, in ns</u>	<u>1 = Inner FEC or ER1 FEC provides information on transmit path data delay with ns resolution in registers 1.1813 through 1.1816</u> <u>0 = Inner FEC or ER1 FEC does not provide information on transmit path data delay with ns resolution</u>	<u>RO</u>
<u>1.1800.5</u>	<u>TimeSync Inner FEC or ER1 FEC receive path data delay ability, in sub-ns</u>	<u>1 = Inner FEC or ER1 FEC provides information on receive path data delay with sub-ns resolution in registers 1.1823 and 1.1824</u> <u>0 = Inner FEC or ER1 FEC does not provide information on receive path data delay with sub-ns resolution</u>	<u>RO</u>
<u>1.1800.4</u>	<u>TimeSync Inner FEC or ER1 FEC receive path data delay ability, in ns</u>	<u>1 = Inner FEC or ER1 FEC provides information on receive path data delay with ns resolution in registers 1.1819 through 1.1823</u> <u>0 = Inner FEC or ER1 FEC does not provide information on receive path data delay with ns resolution</u>	<u>RO</u>
1.1800.3	TimeSync transmit path data delay ability, in sub-ns	1 = PMA/PMD provides information on transmit path data delay with sub-ns resolution in registers 1.1809 and 1.1810 0 = PMA/PMD does not provide information on transmit path data delay with sub-ns resolution	RO
1.1800.2	TimeSync receive path data delay ability, in sub-ns	1 = PMA/PMD provides information on receive path data delay with sub-ns resolution in registers 1.1811 and 1.1812 0 = PMA/PMD does not provide information on receive path data delay with sub-ns resolution	RO

Table 45–139—TimeSync FEC/PMA/PMD capability (*continued*)

Bit(s)	Name	Description	R/W ^a
1.1800.1	TimeSync transmit path data delay ability, in ns	1 = PMA/PMD provides information on transmit path data delay with ns resolution in registers 1.1801 through 1.1804 0 = PMA/PMD does not provide information on transmit path data delay with ns resolution	RO
1.1800.0	TimeSync receive path data delay ability, in ns	1 = PMA/PMD provides information on receive path data delay with ns resolution in registers 1.1805 through 1.1808 0 = PMA/PMD does not provide information on receive path data delay with ns resolution	RO

^a RO = Read only.

Insert 45.2.1.177a through 45.2.1.177c after 45.2.1.177 as follows:

45.2.1.177a TimeSync FEC sublayer transmit path data delay (Registers 1.1813 through 1.1818)

The TimeSync FEC sublayer transmit path data delay registers are used with Inner FEC sublayer and the ER1 FEC sublayer. The TimeSync FEC sublayer transmit path data delay registers contain the maximum and minimum values of the FEC sublayer transmit path data delay. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4), and when the corresponding capability register bit (see Table 45–139) is asserted.

The maximum FEC sublayer transmit path data delay value is provided in two components. The first component (registers 1.1813 and 1.1814, see Table 45–141a) provides the integer nanoseconds portion of the maximum FEC sublayer transmit path data delay, in units of nanoseconds. The second component (register 1.1817, see Table 45–141a) provides the fractional nanoseconds portion of the maximum FEC sublayer transmit path data delay, in units of 2^{-16} ns. If both components are valid (registers 1.1800.6 and 1.1800.7, see Table 45–139), the maximum FEC sublayer transmit path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum FEC sublayer transmit path data delay.

The minimum FEC sublayer transmit path data delay value is provided in two components. The first component (registers 1.1815 and 1.1816, see Table 45–141a) provides the integer nanoseconds portion of the minimum FEC sublayer transmit path data delay, in units of nanoseconds. The second component (register 1.1818, see Table 45–141a) provides the fractional nanoseconds portion of the minimum FEC sublayer transmit path data delay, in units of 2^{-16} ns. If both components are valid (registers 1.1800.6 and 1.1800.7, see Table 45–139), the minimum FEC sublayer transmit path data delay is the sum of the values

from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum FEC sublayer transmit path data delay.

Table 45–141a—TimeSync FEC sublayer transmit path data delay register

Bit(s)	Name	Description	R/W ^a
1.1813.15:0	Maximum FEC sublayer transmit path data delay in ns, lower	FEC_delay_ns_TX_max [15:0]	RO, MW
1.1814.15:0	Maximum FEC sublayer transmit path data delay in ns, upper	FEC_delay_ns_TX_max [31:16]	RO, MW
1.1815.15:0	Minimum FEC sublayer transmit path data delay in ns, lower	FEC_delay_ns_TX_min [15:0]	RO, MW
1.1816.15:0	Minimum FEC sublayer transmit path data delay in ns, upper	FEC_delay_ns_TX_min [31:16]	RO, MW
1.1817.15:0	Maximum FEC sublayer transmit path data delay in sub-ns	FEC_delay_subns_TX_max [15:0]	RO
1.1818.15:0	Minimum FEC sublayer transmit path data delay in sub-ns	FEC_delay_subns_TX_min [15:0]	RO

^a RO = Read only, MW = Multi-word.

45.2.1.177b TimeSync FEC sublayer receive path data delay (Registers 1.1819 through 1.1824)

The TimeSync FEC sublayer receive path data delay registers are used with Inner FEC sublayer and the ER1 FEC sublayer. The TimeSync FEC sublayer receive path data delay registers contain the maximum and minimum values of the FEC sublayer receive path data delay. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4), and when the corresponding capability register bit (see Table 45–139) is asserted.

The maximum FEC sublayer receive path data delay value is provided in two components. The first component (registers 1.1819 and 1.1820, see Table 45–141b) provides the integer nanoseconds portion of the maximum FEC sublayer receive path data delay, in units of nanoseconds. The second component (register 1.1823, see Table 45–141b) provides the fractional nanoseconds portion of the maximum FEC sublayer receive path data delay, in units of 2^{-16} ns. If both components are valid (registers 1.1800.4 and 1.1800.5, see Table 45–139), the maximum FEC sublayer receive path data delay is the sum of the values from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the maximum FEC sublayer receive path data delay.

The minimum FEC sublayer receive path data delay value is provided in two components. The first component (registers 1.1821 and 1.1822, see Table 45–141b) provides the integer nanoseconds portion of the minimum FEC sublayer receive path data delay, in units of nanoseconds. The second component (register 1.1824, see Table 45–141b) provides the fractional nanoseconds portion of the minimum FEC sublayer receive path data delay, in units of 2^{-16} ns. If both components are valid (registers 1.1800.4 and 1.1800.5, see Table 45–139), the minimum FEC sublayer receive path data delay is the sum of the values

from these two components. If either of the two components is not valid, then the value corresponding to the invalid component is not included in the minimum FEC sublayer receive path data delay.

Table 45–141b—TimeSync FEC sublayer receive path data delay register

Bit(s)	Name	Description	R/W ^a
1.1819.15:0	Maximum FEC sublayer receive path data delay in ns, lower	FEC_delay_ns_RX_max [15:0]	RO, MW
1.1820.15:0	Maximum FEC sublayer receive path data delay in ns, upper	FEC_delay_ns_RX_max [31:16]	RO, MW
1.1821.15:0	Minimum FEC sublayer receive path data delay in ns, lower	FEC_delay_ns_RX_min [15:0]	RO, MW
1.1822.15:0	Minimum FEC sublayer receive path data delay in ns, upper	FEC_delay_ns_RX_min [31:16]	RO, MW
1.1823.15:0	Maximum FEC sublayer receive path data delay in sub-ns	FEC_delay_subns_RX_max [15:0]	RO
1.1824.15:0	Minimum FEC sublayer receive path data delay in sub-ns	FEC_delay_subns_RX_min [15:0]	RO

^a RO = Read only, MW = Multi-word.

45.2.1.177c TimeSync FEC configuration (Register 1.1825)

The assignment of bits in the TimeSync FEC configuration register is shown in Table 45–141c.

Table 45–141c—TimeSync FEC configuration

Bit(s)	Name	Description	R/W ^a
1.1825.15	Reserved	Value always 0	RO
1.1825.14	alignment marker location enable	0 = alignment marker location is not enabled 1 = alignment marker location is enabled	R/W
1.1825.13:0	Reserved	Value always 0	RO

^a RO = Read only, R/W = Read/Write.

45.2.1.177c.1 FEC alignment marker location enable (1.1825.14)

Bit 1.1825.14 enables and disables alignment marker location, see 186.2.3.1.5.

45.2.1.216 IFEC control register (Register 1.2200)

Change Table 45–180 (as amended by IEEE Std 802.3ck-2022) as follows:

Table 45–180—IFEC control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2200.15:7 1.2200.15:3	Reserved	Value always 0	RO
1.2200.6	ER1 FEC remote loopback	1 = Enable ER1 FEC remote loopback mode 0 = Disable ER1 FEC remote loopback mode	R/W
1.2200.5	IFEC local loopback	1 = Enable IFEC local loopback mode 0 = Disable IFEC local loopback mode	R/W
1.2200.4	IFEC degraded SER enable	1 = IFEC decoder indicates degraded SER 0 = IFEC decoder does not indicate degraded SER	R/W
1.2200.3	Reserved	Value always 0	RO
1.2200.2	IFEC enable	1 = Inverse RS-FEC is enabled 0 = Inverse RS-FEC is disabled	R/W
1.2200.1	IFEC bypass indication enable	1 = IFEC decoder does not indicate errors 0 = IFEC decoder indicates errors	R/W
1.2200.0	IFEC bypass correction enable	1 = IFEC decoder performs error detection without error correction 0 = IFEC decoder performs error detection and error correction	R/W

^a R/W = Read/Write, RO = Read only.

Insert 45.2.1.216.aa to 45.2.1.216.ac before 45.2.1.216.a (as inserted by IEEE Std 802.3ck-2022) as follows:

45.2.1.216.aa ER1 FEC remote loopback (1.2200.6)

When bit 1.2200.6 is set to one received data is looped back to the transmit path as shown in Figure 186–12.

45.2.1.216.ab IFEC local loopback (1.2200.5)

When bit 1.2200.5 is set to one Inverse RS-FEC decoded data in the transmit direction is looped back to the Inverse RS-FEC encoder in the receive direction.

45.2.1.216.ac IFEC degraded SER enable (1.2200.4)

Bit 1.2200.4 enables the IFEC decoder to indicate the presence of a degraded SER when the ability is supported. When set to a one, this variable enables degraded SER detection. When set to a zero, degraded SER detection is disabled. Writes to this bit are ignored and reads return a zero if the IFEC does not have the ability to signal the presence of a degraded SER.

45.2.1.217 IFEC status register (Register 1.2201)

Change Table 45–181 as follows:

Table 45–181—IFEC status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2201.15	PCS align status	1 = IFEC encoder has locked and aligned all PCS lanes 0 = IFEC encoder has not locked and aligned all PCS lanes	RO
1.2201.14	IFEC align status	1 = IFEC receive lanes locked and aligned 0 = IFEC receive lanes not locked and aligned	RO
1.2201.13+12	Reserved	Value always 0	RO
<u>1.2201.12</u>	<u>IFEC degraded SER</u>	<u>1 = IFEC SER is degraded</u> <u>0 = IFEC SER is not degraded</u>	<u>RO</u>
1.2201.11	IFEC AM lock 3	1 = IFEC lane 3 locked and aligned 0 = IFEC lane 3 not locked and aligned	RO
1.2201.10	IFEC AM lock 2	1 = IFEC lane 2 locked and aligned 0 = IFEC lane 2 not locked and aligned	RO
1.2201.9	IFEC AM lock 1	1 = IFEC lane 1 locked and aligned 0 = IFEC lane 1 not locked and aligned	RO
1.2201.8	IFEC AM lock 0	1 = IFEC lane 0 locked and aligned 0 = IFEC lane 0 not locked and aligned	RO
1.2201.7:3	Reserved	Value always 0	RO
<u>1.2201.7</u>	<u>ER1 FEC remote loopback ability</u>	<u>1 = ER1 FEC has the ability to perform remote loopback</u> <u>0 = ER1 FEC does not have the ability to perform remote loopback</u>	<u>RO</u>
<u>1.2201.6</u>	<u>IFEC local loopback ability</u>	<u>1 = IFEC decoder has the ability to perform local loopback</u> <u>0 = IFEC decoder does not have the ability to perform local loopback</u>	<u>RO</u>
<u>1.2201.5</u>	<u>Received local degraded</u>	<u>1 = IFEC has received local degraded status indication</u> <u>0 = IFEC has not received local degraded status indication</u>	<u>RO</u>
<u>1.2201.4</u>	<u>Received remote degraded</u>	<u>1 = IFEC has received remote degraded status indication</u> <u>0 = IFEC has not received remote degraded status</u>	<u>RO</u>
<u>1.2201.3</u>	<u>IFEC degraded SER ability</u>	<u>1 = IFEC decoder has the ability to indicate the presence of a degraded SER</u> <u>0 = IFEC decoder does not have the ability to indicate the presence of a degraded SER</u>	<u>RO</u>
1.2201.2	IFEC high SER	1 = IFEC errors have exceeded threshold 0 = IFEC errors have not exceeded threshold	RO/LH
1.2201.1	IFEC bypass indication ability	1 = IFEC decoder has the ability to bypass error indication 0 = IFEC decoder does not have the ability to bypass error indication	RO
1.2201.0	IFEC bypass correction ability	1 = IFEC decoder has the ability to bypass error correction 0 = IFEC decoder does not have the ability to bypass error correction	RO

^a RO = Read only, LH = Latching high.

Insert 45.2.1.217.2a after 45.2.1.217.2 as follows:

45.2.1.217.2a IFEC degraded SER (1.2201.12)

When read one, bit 1.2201.12 indicates the IFEC has detected a degraded received signal.

Insert 45.2.1.217.6a through 45.2.1.217.6d after 45.2.1.217.6 as follows:

45.2.1.217.6a 800GBASE-ER1 FEC remote loopback ability (1.2201.7)

When bit 1.2201.7 is set to one the 800GBASE-ER1 FEC has the ability to loop back receive data to the transmit path as shown in Figure 186–12.

45.2.1.217.6b IFEC local loopback ability (1.2201.6)

When bit 1.2201.6 is set to one there is the ability for the Inverse RS-FEC decoded data in the transmit direction to be looped back to the Inverse RS-FEC encoder in the receive direction.

45.2.1.217.6c IFEC received local degraded (1.2201.5)

When read as one, bit 1.2201.5 indicates that the local degraded SER signal has been received. This bit reflects the state of rx_local_degraded for the Inverse RS-FEC.

45.2.1.217.6d IFEC received remote degraded (1.2201.4)

When read as one, bit 1.2201.4 indicates that the remote degraded SER signal has been received. This bit reflects the state of rx_rm_degraded for the Inverse RS-FEC.

45.2.1.217.6e IFEC degraded SER ability (1.2201.3)

This bit is set to one to indicate that the decoder has the ability to indicate the presence of a degraded SER. The bit is set to zero if this ability is not supported.

Insert 45.2.1.220a after 45.2.1.220 as follows:

45.2.1.220a IFEC codeword counter register (Register 1.2207, 1.2208, 1.2209)

The assignment of bits in the IFEC codeword counter register is shown in Table 45–184a. It is a 48-bit counter that counts once for each IFEC codeword received when align status is set to one. Its bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Registers 1.2207, 1.2209, and 1.2209 are used to read the value of the 48-bit counter. When using these registers to read the 48-bit counter value, the register 1.2207 is read first, the values of the registers 1.2208

and 1.2209 are latched when (and only when) register 1.2207 is read, and reads of registers 1.2208 and 1.2209 return their latched values rather than the current value of the counter.

Table 45–184a—IFEC codeword counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.2207.15:0	IFEC codewords lower	IFEC_cw_counter[15:0]	RO, NR
1.2208.15:0	IFEC codewords middle	IFEC_cw_counter[31:16]	RO, NR
1.2209.15:0	IFEC codewords upper	IFEC_cw_counter[47:32]	RO, NR

^a RO = Read only, NR = Non Roll-over.

Change 45.2.1.222 as follows:

45.2.1.222 IFEC symbol error counter, lane 1 through 31 3 (Register 1.2212, ~~1.2213, 1.2214, 1.2215, 1.2216, 1.2217~~ through 1.2241, and 1.2770 through 1.2801)

The behavior of the IFEC symbol error counters, lane 1 through 31 3 is identical to that described for FEC lane 0 in 45.2.1.221. Errors detected in each FEC lane are counted and shown in the corresponding register. Registers 1.2212 through 1.2241 are used for lanes 1 through 15, and registers 1.2770 through 1.2801 are used for lanes 16 through 31. FEC lane 1, lower 16 bits are shown in register 1.2212; FEC lane 1, upper 16 bits are shown in register 1.2213; FEC lane 2, lower 16 bits are shown in register 1.2214; FEC lane 2, upper 16 bits are shown in register 1.2215; etc through register 1.2217 for FEC lane 3, upper 16 bits.

Insert 45.2.1.251 through 45.2.1.273 after 45.2.1.250 as follows:

45.2.1.251 Inner FEC PRBS13Q testing control register (Register 1.2401)

The Inner FEC PRBS13Q testing control register is used for PHY types that implement PRBS13Q testing in the Inner FEC. The assignment of bits in the PRBS13Q testing control register is shown in Table 45–212a.

Table 45–212a—Inner FEC PRBS13Q testing control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2401.15:8	Reserved	Value always 0	RO
1.2401.7	Lane 7 PRBS13Q enable	1 = Enable PRBS13Q on lane 7 0 = Disable PRBS13Q on lane 7	R/W
1.2401.6	Lane 6 PRBS13Q enable	1 = Enable PRBS13Q on lane 6 0 = Disable PRBS13Q on lane 6	R/W
1.2401.5	Lane 5 PRBS13Q enable	1 = Enable PRBS13Q on lane 5 0 = Disable PRBS13Q on lane 5	R/W

Table 45–212a—Inner FEC PRBS13Q testing control register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.2401.4	Lane 4 PRBS13Q enable	1 = Enable PRBS13Q on lane 4 0 = Disable PRBS13Q on lane 4	R/W
1.2401.3	Lane 3 PRBS13Q enable	1 = Enable PRBS13Q on lane 3 0 = Disable PRBS13Q on lane 3	R/W
1.2401.2	Lane 2 PRBS13Q enable	1 = Enable PRBS13Q on lane 2 0 = Disable PRBS13Q on lane 2	R/W
1.2401.1	Lane 1 PRBS13Q enable	1 = Enable PRBS13Q on lane 1 0 = Disable PRBS13Q on lane 1	R/W
1.2401.0	Lane 0 PRBS13Q enable	1 = Enable PRBS13Q on lane 0 0 = Disable PRBS13Q on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.252 Inner FEC PRBS31 testing control register (Register 1.2402)

The Inner FEC PRBS31 testing control register is used for PHY types that implement PRBS31 testing in the Inner FEC. The assignment of bits in the PRBS31 testing control register is shown in Table 45–212b.

Table 45–212b—Inner FEC PRBS31 testing control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2402.15:8	Reserved	Value always 0	RO
1.2402.7	Lane 7 PRBS31 enable	1 = Enable PRBS31 on lane 7 0 = Disable PRBS31 on lane 7	R/W
1.2402.6	Lane 6 PRBS31 enable	1 = Enable PRBS31 on lane 6 0 = Disable PRBS31 on lane 6	R/W
1.2402.5	Lane 5 PRBS31 enable	1 = Enable PRBS31 on lane 5 0 = Disable PRBS31 on lane 5	R/W
1.2402.4	Lane 4 PRBS31 enable	1 = Enable PRBS31 on lane 4 0 = Disable PRBS31 on lane 4	R/W
1.2402.3	Lane 3 PRBS31 enable	1 = Enable PRBS31 on lane 3 0 = Disable PRBS31 on lane 3	R/W
1.2402.2	Lane 2 PRBS31 enable	1 = Enable PRBS31 on lane 2 0 = Disable PRBS31 on lane 2	R/W
1.2402.1	Lane 1 PRBS31 enable	1 = Enable PRBS31 on lane 1 0 = Disable PRBS31 on lane 1	R/W
1.2402.0	Lane 0 PRBS31 enable	1 = Enable PRBS31 on lane 0 0 = Disable PRBS31 on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.253 Inner FEC SSPRQ testing control register (Register 1.2403)

The Inner FEC SSPRQ testing control register is used for PHY types that implement SSPRQ testing in the Inner FEC. The assignment of bits in the SSPRQ testing control register is shown in Table 45–212c.

Table 45–212c—Inner FEC SSPRQ testing control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2403.15:8	Reserved	Value always 0	RO
1.2403.7	Lane 7 SSPRQ enable	1 = Enable SSPRQ on lane 7 0 = Disable SSPRQ on lane 7	R/W
1.2403.6	Lane 6 SSPRQ enable	1 = Enable SSPRQ on lane 6 0 = Disable SSPRQ on lane 6	R/W
1.2403.5	Lane 5 SSPRQ enable	1 = Enable SSPRQ on lane 5 0 = Disable SSPRQ on lane 5	R/W
1.2403.4	Lane 4 SSPRQ enable	1 = Enable SSPRQ on lane 4 0 = Disable SSPRQ on lane 4	R/W
1.2403.3	Lane 3 SSPRQ enable	1 = Enable SSPRQ on lane 3 0 = Disable SSPRQ on lane 3	R/W
1.2403.2	Lane 2 SSPRQ enable	1 = Enable SSPRQ on lane 2 0 = Disable SSPRQ on lane 2	R/W
1.2403.1	Lane 1 SSPRQ enable	1 = Enable SSPRQ on lane 1 0 = Disable SSPRQ on lane 1	R/W
1.2403.0	Lane 0 SSPRQ enable	1 = Enable SSPRQ on lane 0 0 = Disable SSPRQ on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.254 Inner FEC Square Wave testing control register (Register 1.2404)

The Inner FEC Square Wave testing control register is used for PHY types that implement Square Wave testing in the Inner FEC. The assignment of bits in the Square Wave testing control register is shown in Table 45–212d.

Table 45–212d—Inner FEC Square Wave testing control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2404.15:8	Reserved	Value always 0	RO
1.2404.7	Lane 7 Square Wave enable	1 = Enable Square Wave on lane 7 0 = Disable Square Wave on lane 7	R/W
1.2404.6	Lane 6 Square Wave enable	1 = Enable Square Wave on lane 6 0 = Disable Square Wave on lane 6	R/W

Table 45–212d—Inner FEC Square Wave testing control register bit definitions (*continued*)

Bit(s)	Name	Description	R/W ^a
1.2404.5	Lane 5 Square Wave enable	1 = Enable Square Wave on lane 5 0 = Disable Square Wave on lane 5	R/W
1.2404.4	Lane 4 Square Wave enable	1 = Enable Square Wave on lane 4 0 = Disable Square Wave on lane 4	R/W
1.2404.3	Lane 3 Square Wave enable	1 = Enable Square Wave on lane 3 0 = Disable Square Wave on lane 3	R/W
1.2404.2	Lane 2 Square Wave enable	1 = Enable Square Wave on lane 2 0 = Disable Square Wave on lane 2	R/W
1.2404.1	Lane 1 Square Wave enable	1 = Enable Square Wave on lane 1 0 = Disable Square Wave on lane 1	R/W
1.2404.0	Lane 0 Square Wave enable	1 = Enable Square Wave on lane 0 0 = Disable Square Wave on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.255 PRBS31 encoded by Inner FEC control register (Register 1.2405)

The PRBS31 encoded by Inner FEC control register is used to enable PRBS31 test patterns from the PMA layer above the Inner FEC on a per lane basis. The assignment of bits in the PRBS31 encoded by Inner FEC control register is shown in Table 45–212e.

Table 45–212e—PRBS31 encoded by Inner FEC control register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2405.15:8	Reserved	Value always 0	RO
1.2405.7	Lane 7 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 7 0 = Disable PRBS31 encoded by Inner FEC on lane 7	R/W
1.2405.6	Lane 6 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 6 0 = Disable PRBS31 encoded by Inner FEC on lane 6	R/W
1.2405.5	Lane 5 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 5 0 = Disable PRBS31 encoded by Inner FEC on lane 5	R/W
1.2405.4	Lane 4 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 4 0 = Disable PRBS31 encoded by Inner FEC on lane 4	R/W
1.2405.3	Lane 3 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 3 0 = Disable PRBS31 encoded by Inner FEC on lane 3	R/W
1.2405.2	Lane 2 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 2 0 = Disable PRBS31 encoded by Inner FEC on lane 2	R/W
1.2405.1	Lane 1 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 1 0 = Disable PRBS31 encoded by Inner FEC on lane 1	R/W
1.2405.0	Lane 0 PRBS31 encoded by Inner FEC enable	1 = Enable PRBS31 encoded by Inner FEC on lane 0 0 = Disable PRBS31 encoded by Inner FEC on lane 0	R/W

^a R/W = Read/Write, RO = Read only.

45.2.1.256 Inner FEC alignment status register (Register 1.2410)

The assignment of bits in the Inner FEC alignment status register is shown in Table 45–212f.

Table 45–212f—Inner FEC alignment status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2410.15	Inner FEC alignment 7	1 = Lane 7 transmit direction is aligned 0 = Lane 7 transmit direction is not aligned	RO
1.2410.14	Inner FEC alignment 6	1 = Lane 6 transmit direction is aligned 0 = Lane 6 transmit direction is not aligned	RO
1.2410.13	Inner FEC alignment 5	1 = Lane 5 transmit direction is aligned 0 = Lane 5 transmit direction is not aligned	RO
1.2410.12	Inner FEC alignment 4	1 = Lane 4 transmit direction is aligned 0 = Lane 4 transmit direction is not aligned	RO
1.2410.11	Inner FEC alignment 3	1 = Lane 3 transmit direction is aligned 0 = Lane 3 transmit direction is not aligned	RO
1.2410.10	Inner FEC alignment 2	1 = Lane 2 transmit direction is aligned 0 = Lane 2 transmit direction is not aligned	RO
1.2410.9	Inner FEC alignment 1	1 = Lane 1 transmit direction is aligned 0 = Lane 1 transmit direction is not aligned	RO
1.2410.8	Inner FEC alignment 0	1 = Lane 0 transmit direction is aligned 0 = Lane 0 transmit direction is not aligned	RO
1.2410.7	Inner FEC lock 7	1 = Lane 7 receive direction is locked 0 = Lane 7 receive direction is not locked	RO
1.2410.6	Inner FEC lock 6	1 = Lane 6 receive direction is locked 0 = Lane 6 receive direction is not locked	RO
1.2410.5	Inner FEC lock 5	1 = Lane 5 receive direction is locked 0 = Lane 5 receive direction is not locked	RO
1.2410.4	Inner FEC lock 4	1 = Lane 4 receive direction is locked 0 = Lane 4 receive direction is not locked	RO
1.2410.3	Inner FEC lock 3	1 = Lane 3 receive direction is locked 0 = Lane 3 receive direction is not locked	RO
1.2410.2	Inner FEC lock 2	1 = Lane 2 receive direction is locked 0 = Lane 2 receive direction is not locked	RO
1.2410.1	Inner FEC lock 1	1 = Lane 1 receive direction is locked 0 = Lane 1 receive direction is not locked	RO
1.2410.0	Inner FEC lock 0	1 = Lane 0 receive direction is locked 0 = Lane 0 receive direction is not locked	RO

^a RO = Read only.

45.2.1.257 Inner FEC status 1 register (Register 1.2411)

The assignment of bits in the Inner FEC status 1 register is shown in Table 45–212g. All the bits in the Inner FEC status 1 register are read only; therefore, a write to the Inner FEC status 1 register shall have no effect.

Table 45–212g—Inner FEC status 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2411.15:3	Reserved	Value always 0	RO
1.2411.2	Remote PHY fault indication	Set to 1 when the 800GBASE-ER1 FEC receive function has detected remote fault indicated in the status octet (see 186.2.3.5.2)	RO
1.2411.1:0	DSP lock	Set to 1 when the 800GBASE-LR1 Inner FEC receive function has detected the location of the pilot sequence for a given polarization symbol stream (see 184.8.2.1)	RO

^a RO = Read only, LL = Latching low.

45.2.1.258 Inner or ER1 FEC lane 0 corrected codewords counter (Register 1.2412, 1.2413)

Table 45–212h shows the assignment of bits for the lane 0 Inner or ER1 FEC corrected codewords counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2412 and 1.2413 are used to read the value of a 32-bit counter. When registers 1.2412 and 1.2413 are used to read the 32-bit counter value, register 1.2412 is read first, the value of the register 1.2413 is latched when (and only when) register 1.2412 is read, and reads of register 1.2413 return the latched value rather than the current value of the counter.

Table 45–212h—Inner or ER1 FEC lane 0 corrected codewords counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.2412.15:0	Inner FEC or ER1 FEC corrected codewords lower	FEC_corrected_cw_counter[15:0]	RO, NR
1.2413.15:0	Inner FEC or ER1 FEC corrected codewords upper	FEC_corrected_cw_counter[31:16]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.259 Inner or ER1 FEC lane 0 uncorrected codewords counter (Register 1.2414, 1.2415)

Table 45–212i shows the assignment of bits for the lane 0 Inner or ER1 FEC uncorrected codewords counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2414 and 1.2415 are used to read the value of a 32-bit counter. When registers 1.2414 and 1.2415 are used to read the 32-bit counter value, register 1.2414 is read first, the value of the register 1.2415 is latched when (and only when) register

1.2414 is read, and reads of register 1.2415 return the latched value rather than the current value of the counter.

Table 45–212i—Inner or ER1 FEC lane 0 uncorrected codewords counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.2414.15:0	Inner FEC or ER1 FEC uncorrected codewords lower	FEC_uncorrected_cw_counter[15:0]	RO, NR
1.2415.15:0	Inner FEC or ER1 FEC uncorrected codewords upper	FEC_uncorrected_cw_counter[31:16]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.260 Inner or ER1 FEC lane 0 total bits register (Register 1.2416, 1.2417, 1.2418, 1.2419)

Table 45–212j shows the assignment of bits for the lane 0 Inner or ER1 FEC total bits register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2416, 1.2417, 1.2418, and 1.2419 are used to read the value of a 64-bit counter. When registers 1.2416, 1.2417, 1.2418, and 1.2419 are used to read the 64-bit counter value, register 1.2416 is read first, the values of registers 1.2417, 1.2418, and 1.2419 are latched when (and only when) register 1.2416 is read, and reads of registers 1.2417, 1.2418, and 1.2419 return the latched value rather than the current value of the counter.

Table 45–212j—Inner or ER1 FEC lane 0 total bits register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2416.15:0	Inner FEC or ER1 FEC total bits lowest	FEC_total_bits[15:0]	RO, NR
1.2417.15:0	Inner FEC or ER1 FEC total bits	FEC_total_bits[31:16]	RO, NR
1.2418.15:0	Inner FEC or ER1 FEC total bits	FEC_total_bits[47:32]	RO, NR
1.2419.15:0	Inner FEC or ER1 FEC total bits highest	FEC_total_bits[64:48]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.261 Inner or ER1 FEC lane 0 corrected bits register (Register 1.2420, 1.2421, 1.2422, 1.2423)

Table 45–212k shows the assignment of bits for the lane 0 Inner or ER1 FEC corrected bits register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2420, 1.2421, 1.2422, and 1.2423 are used to read the value of a 64-bit counter. When registers 1.2420, 1.2421, 1.2422, and 1.2423 are used to read the 64-bit counter value, register 1.2420 is read first, the values of registers 1.2421, 1.2422, and 1.2423

are latched when (and only when) register 1.2420 is read, and reads of registers 1.2421, 1.2422, and 1.2423 return the latched value rather than the current value of the counter.

Table 45–212k—Inner or ER1 lane 0 FEC corrected bits register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2420.15:0	Inner FEC or ER1 FEC corrected bits lowest	FEC_corrected_bits[15:0]	RO, NR
1.2421.15:0	Inner FEC or ER1 FEC corrected bits	FEC_corrected_bits[31:16]	RO, NR
1.2422.15:0	Inner FEC or ER1 FEC corrected bits	FEC_corrected_bits[47:32]	RO, NR
1.2423.15:0	Inner FEC or ER1 FEC corrected bits highest	FEC_corrected_bits[64:48]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.262 Inner FEC lane 0 codeword counter register (Register 1.2424, 1.2425, 1.2426)

The assignment of bits in the Inner FEC codeword counter register is shown in Table 45–212l. It is a 48-bit counter that counts once for each Inner FEC codeword received when align status is set to one. Its bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Registers 1.2424, 1.2425, and 1.2426 are used to read the value of the 48-bit counter. When using these registers to read the 48-bit counter value, the register 1.2424 is read first, the values of the registers 1.2425 and 1.2426 are latched when (and only when) register 1.2424 is read, and reads of registers 1.2425 and 1.2426 return their latched values rather than the current value of the counter.

Table 45–212l—Inner FEC lane 0 codeword counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.2424.15:0	Inner FEC codewords lower	Inner_FEC_cw_counter[15:0]	RO, NR
1.2425.15:0	Inner FEC codewords middle	Inner_FEC_cw_counter[31:16]	RO, NR
1.2426.15:0	Inner FEC codewords upper	Inner_FEC_cw_counter[47:32]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.263 Inner FEC lane 0 codeword error bin registers 0 through 4 (Registers 1.2427 through 1.2436)

Table 45–212m shows the assignment of bits for the Inner FEC lane 0 codeword error bin registers. The Inner FEC codeword error bin registers increment depending upon the error signature of a corrected codeword. Their bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Two registers are used to read the value of each 32-bit counter, the value of the second register is latched when the first register is read.

The bin 0 register (1.2427, 1.2428) keeps a count of codewords with no bit errors, the bin 1 register (1.2429, 1.2430) keeps a count of codewords with 1 bit error corrected, and the bin 2 register (1.2431, 1.2432) keeps a count of codewords with 2 bits corrected.

For the Inner FEC defined in Clause 184, the bin 3 (1.2433, 1.2434) register keeps a count of codewords with 3 bits corrected, and the bin 4 (1.2435, 1.2436) register keeps a count of codewords with 4 or more bits corrected.

The Inner FEC defined in Clause 177 does not use the bin 4 register, for Clause 177 the bin 3 register keeps a count of codewords with 3 or more bits corrected.

Table 45–212m—Inner FEC lane 0 codeword error bin register definitions

Bit(s)	Name	Description	R/W ^a
1.2427.15:0	Inner FEC codeword error bin 0 lower	Inner_FEC_codeword_error_bin_0[15:0]	RO, NR
1.2428.15:0	Inner FEC codeword error bin 0 upper	Inner_FEC_codeword_error_bin_0[31:16]	RO, NR
1.2429.15:0	Inner FEC codeword error bin 1 lower	Inner_FEC_codeword_error_bin_1[15:0]	RO, NR
1.2430.15:0	Inner FEC codeword error bin 1 upper	Inner_FEC_codeword_error_bin_1[31:16]	RO, NR
1.2431.15:0	Inner FEC codeword error bin 2 lower	Inner_FEC_codeword_error_bin_2[15:0]	RO, NR
1.2432.15:0	Inner FEC codeword error bin 2 upper	Inner_FEC_codeword_error_bin_2[31:16]	RO, NR
1.2433.15:0	Inner FEC codeword error bin 3 lower	Inner_FEC_codeword_error_bin_3[15:0]	RO, NR
1.2434.15:0	Inner FEC codeword error bin 3 upper	Inner_FEC_codeword_error_bin_3[31:16]	RO, NR
1.2435.15:0	Inner FEC codeword error bin 4 lower	Inner_FEC_codeword_error_bin_4[15:0]	RO, NR
1.2436.15:0	Inner FEC codeword error bin 4 upper	Inner_FEC_codeword_error_bin_4[31:16]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.264 Inner FEC counter registers for lanes 1 through 7 (Registers 1.2437 through 1.2611)

The twenty five counter registers 1.2412 through 1.2436 are repeated for each Inner FEC lane present, with registers 1.2412 through 1.2436 being for lane 0, registers 1.2437 through 1.2461 being for lane 1, etc.

45.2.1.265 ER1 FEC CRC32 blocks processed counter register (Register 1.2640, 1.2641, 1.2642)

The assignment of bits in the ER1 FEC CRC32 blocks processed counter register is shown in Table 45–212n. It is a 48-bit counter that counts once for each CRC32 block processed. These bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Registers 1.2640, 1.2641, and 1.2642 are used to read the value of the 48-bit counter. When using these registers to read the 48-bit counter value, the register 1.2640 is read first, the values of the registers 1.2641

and 1.2642 are latched when (and only when) register 1.2640 is read, and reads of registers 1.2641 and 1.2642 return their latched values rather than the current value of the counter.

Table 45–212n—ER1 FEC CRC32 blocks processed counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.2640.15:0	CRC32 blocks codewords lower	FEC_CRC_total_blocks_counter[15:0]	RO, NR
1.2641.15:0	CRC32 blocks codewords middle	FEC_CRC_total_blocks_counter[31:16]	RO, NR
1.2642.15:0	CRC32 block codewords upper	FEC_CRC_total_blocks_counter[47:32]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.266 ER1 FEC CRC32 error counter register (Register 1.2643)

The assignment of bits in the ER1 CRC32 error counter register is shown in Table 45–212o. These bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Table 45–212o—ER1 FEC CRC32 error counter bit definitions

Bit(s)	Name	Description	R/W ^a
1.2643.15:0	ER1 FEC CRC32 error counter	FEC_CRC_errored_blocks_counter[15:0]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.267 Inner FEC test block error bin counter registers (Registers 1.2644 through 1.2694)

The Inner FEC test block error bin counter registers provide FEC error statistics from a PRBS data stream. These registers are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow. Three registers are used to read the value of each 48-bit counter. The values of all registers are latched when the first register of bin 0 is read.

There are 17 bin counters. Each 48-bit counter consumes three registers, 17 counters span a total of 51 registers. The bin 0 register keeps a count of test blocks with no test symbol errors. The bin 1 register keeps a count of test blocks with 1 test symbol error. The bin 2 register keeps a count of test blocks with 2 test symbol errors, and so on up to 15 test symbol errors. The bin 16p register counts test blocks with 16 or more test symbol errors.

Table 45–212p shows the assignment of bits for the Inner FEC test block error bin 0 and 1 registers.

Table 45–212p—Inner FEC test block error bin register definitions (for 2 of the 17 bins)

Bit(s)	Name	Description	R/W ^a
1.2644.15:0	Inner FEC test block error bin 0 lower	Inner_FEC_test_block_error_bin_0[15:0]	RO, NR
1.2645.15:0	Inner FEC test block error bin 0 middle	Inner_FEC_test_block_error_bin_0[31:16]	RO, NR
1.2646.15:0	Inner FEC test block error bin 0 upper	Inner_FEC_test_block_error_bin_0[47:32]	RO, NR
1.2647.15:0	Inner FEC test block error bin 1 lower	Inner_FEC_test_block_error_bin_1[15:0]	RO, NR
1.2648.15:0	Inner FEC test block error bin 1 middle	Inner_FEC_test_block_error_bin_1[31:16]	RO, NR
1.2649.15:0	Inner FEC test block error bin 1 upper	Inner_FEC_test_block_error_bin_1[47:32]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.268 IFEC lane 0 mapping register (Register 1.2700)

The assignment of bits in the IFEC lane 0 mapping register is shown in Table 45–212q. When the IFEC detects and locks the RS-FEC for IFEC service interface lane 0, the detected RS-FEC lane number is recorded in this register. The contents of the Lane 0 mapping register is valid when the Inverse RS-FEC align status (1.2201.14) is set to one and are invalid otherwise.

Table 45–212q—IFEC lane 0 mapping register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2700.15:6	Reserved	Value always 0	RO
1.2700.5:0	Lane 0 mapping	PCS lane received in service interface lane 0	RO

^a RO = Read only.

45.2.1.269 IFEC lanes 1 through 31 mapping registers (Registers 1.2701 through 1.2731)

The definition of Lanes 1 through 31 mapping registers is identical to that described for lane 0 in 45.2.1.268. The lane mapping for lane 1 is in register 1.2701; lane 2 is in register 1.2702; etc.

45.2.1.270 IFEC alignment status register (Register 1.2732, 1.2733)

The assignment of bits in the IFEC alignment status register is shown in Table 45–212r and Table 45–212s. All the bits in the IFEC alignment status register are read only; a write to the IFEC PCS alignment status register shall have no effect.

Table 45–212r—IFEC alignment status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2732.15	Lane 15 aligned	1 = Lane 15 alignment marker is locked 0 = Lane 15 alignment marker is not locked	RO
1.2732.14	Lane 14 aligned	1 = Lane 14 alignment marker is locked 0 = Lane 14 alignment marker is not locked	RO
1.2732.13	Lane 13 aligned	1 = Lane 13 alignment marker is locked 0 = Lane 13 alignment marker is not locked	RO
1.2732.12	Lane 12 aligned	1 = Lane 12 alignment marker is locked 0 = Lane 12 alignment marker is not locked	RO
1.2732.11	Lane 11 aligned	1 = Lane 11 alignment marker is locked 0 = Lane 11 alignment marker is not locked	RO
1.2732.10	Lane 10 aligned	1 = Lane 10 alignment marker is locked 0 = Lane 10 alignment marker is not locked	RO
1.2732.9	Lane 9 aligned	1 = Lane 9 alignment marker is locked 0 = Lane 9 alignment marker is not locked	RO
1.2732.8	Lane 8 aligned	1 = Lane 8 alignment marker is locked 0 = Lane 8 alignment marker is not locked	RO
1.2732.7	Lane 7 aligned	1 = Lane 7 alignment marker is locked 0 = Lane 7 alignment marker is not locked	RO
1.2732.6	Lane 6 aligned	1 = Lane 6 alignment marker is locked 0 = Lane 6 alignment marker is not locked	RO
1.2732.5	Lane 5 aligned	1 = Lane 5 alignment marker is locked 0 = Lane 5 alignment marker is not locked	RO
1.2732.4	Lane 4 aligned	1 = Lane 4 alignment marker is locked 0 = Lane 4 alignment marker is not locked	RO
1.2732.3	Lane 3 aligned	1 = Lane 3 alignment marker is locked 0 = Lane 3 alignment marker is not locked	RO
1.2732.2	Lane 2 aligned	1 = Lane 2 alignment marker is locked 0 = Lane 2 alignment marker is not locked	RO
1.2732.1	Lane 1 aligned	1 = Lane 1 alignment marker is locked 0 = Lane 1 alignment marker is not locked	RO
1.2732.0	Lane 0 aligned	1 = Lane 0 alignment marker is locked 0 = Lane 0 alignment marker is not locked	RO

^a RO = Read only.

Table 45–212s—IFEC alignment status register bit definitions

Bit(s)	Name	Description	R/W ^a
1.2733.15	Lane 31 aligned	1 = Lane 31 alignment marker is locked 0 = Lane 31 alignment marker is not locked	RO
1.2733.14	Lane 30 aligned	1 = Lane 30 alignment marker is locked 0 = Lane 30 alignment marker is not locked	RO
1.2733.13	Lane 29 aligned	1 = Lane 29 alignment marker is locked 0 = Lane 29 alignment marker is not locked	RO
1.2733.12	Lane 28 aligned	1 = Lane 28 alignment marker is locked 0 = Lane 28 alignment marker is not locked	RO
1.2733.11	Lane 27 aligned	1 = Lane 27 alignment marker is locked 0 = Lane 27 alignment marker is not locked	RO
1.2733.10	Lane 26 aligned	1 = Lane 26 alignment marker is locked 0 = Lane 26 alignment marker is not locked	RO
1.2733.9	Lane 25 aligned	1 = Lane 25 alignment marker is locked 0 = Lane 25 alignment marker is not locked	RO
1.2733.8	Lane 24 aligned	1 = Lane 24 alignment marker is locked 0 = Lane 24 alignment marker is not locked	RO
1.2733.7	Lane 23 aligned	1 = Lane 23 alignment marker is locked 0 = Lane 23 alignment marker is not locked	RO
1.2733.6	Lane 22 aligned	1 = Lane 22 alignment marker is locked 0 = Lane 22 alignment marker is not locked	RO
1.2733.5	Lane 21 aligned	1 = Lane 21 alignment marker is locked 0 = Lane 21 alignment marker is not locked	RO
1.2733.4	Lane 20 aligned	1 = Lane 20 alignment marker is locked 0 = Lane 20 alignment marker is not locked	RO
1.2733.3	Lane 19 aligned	1 = Lane 19 alignment marker is locked 0 = Lane 19 alignment marker is not locked	RO
1.2733.2	Lane 18 aligned	1 = Lane 18 alignment marker is locked 0 = Lane 18 alignment marker is not locked	RO
1.2733.1	Lane 17 aligned	1 = Lane 17 alignment marker is locked 0 = Lane 17 alignment marker is not locked	RO
1.2733.0	Lane 16 aligned	1 = Lane 16 alignment marker is locked 0 = Lane 16 alignment marker is not locked	RO

^a RO = Read only.

45.2.1.271 IFEC codeword error bin registers 1 through 15 (Registers 1.2734 through 1.2763)

The assignment of bits in the IFEC codeword error bin 1 register is shown in Table 45–212t. The assignment of bits in the other IFEC codeword error bin registers is identical to that of bin 1. The IFEC codeword error bin registers increment depending upon the error signature of a corrected codeword. Their bits are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow.

Two registers are used to read the value of each 32-bit counter, the value of the second register is latched when the first register is read. For example when registers 1.2734 and 1.2735 are used to read the 32-bit counter value of FEC_codeword_error_bin_1, the register 1.2734 is read first, the value of the register 1.2735 is latched when (and only when) register 1.2734 is read, and reads of register 1.2735 return the latched value rather than the current value of the counter.

Table 45–212t—IFEC codeword error bin 1 bit definitions

Bit(s)	Name	Description	R/W ^a
1.2734.15:0	IFEC codeword error bin 1 lower	IFEC_codeword_error_bin_1[15:0]	RO, NR
1.2735.15:0	IFEC codeword error bin 1 upper	IFEC_codeword_error_bin_1[31:16]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.272 PMA test block error bin counter registers (Registers 1.2900 through 1.3715)

The PMA test block error bin counter registers provide emulation of FEC error statistics from a PRBS data stream. These registers are reset to all zeros when the register is read by the management function or upon reset, and held at all ones in the case of overflow. Three registers are used to read the value of each 48-bit counter. The values of all registers for a given PMAL are latched when the first register of bin 0 is read.

There are 17 bin counters for eight PMALs. Each 48-bit counter consumes three registers, so 17 counters for eight PMALs spans a total of 408 registers in each direction. The bin 0 register keeps a count of test blocks with no test symbol errors. The bin 1 register keeps a count of test blocks with 1 test symbol error. The bin 2 register keeps a count of test blocks with 2 test symbol errors, and so on up to 15 test symbol errors. The bin 16p register counts test blocks with 16 or more test symbol errors.

Table 45–212u shows the assignment of bits for the lane 0 PMA test block error bin 0 to bin 3 registers in the transmit direction. Lane 0 error bin registers end at 1.2950 with test_block_error_bin_tx_0_16p upper. Lane 1 error bin registers start at 1.2951 with test_block_error_bin_tx_1_0 lower. The transmit direction error bin counter registers span from 1.2900 to 1.3307. Table 45–212v shows the assignment of bits for the lane 0 PMA test block error bin 0 to bin 3 registers in the receive direction. The receive direction registers span from 1.3308 to 1.3715.

Table 45–212u—Lane 0 PMA transmit test block error bin register definitions (for 4 of the 17 bins)

Bit(s)	Name	Description	R/W ^a
1.2900.15:0	PMA transmit test block error bin 0 lower	test_block_error_bin_tx_0_0[15:0]	RO, NR
1.2901.15:0	PMA transmit test block error bin 0 middle	test_block_error_bin_tx_0_0[31:16]	RO, NR
1.2902.15:0	PMA transmit test block error bin 0 upper	test_block_error_bin_tx_0_0[47:32]	RO, NR
1.2903.15:0	PMA transmit test block error bin 1 lower	test_block_error_bin_tx_0_1[15:0]	RO, NR
1.2904.15:0	PMA transmit test block error bin 1 middle	test_block_error_bin_tx_0_1[31:16]	RO, NR
1.2905.15:0	PMA transmit test block error bin 1 upper	test_block_error_bin_tx_0_1[47:32]	RO, NR
1.2906.15:0	PMA transmit test block error bin 2 lower	test_block_error_bin_tx_0_2[15:0]	RO, NR
1.2907.15:0	PMA transmit test block error bin 2 middle	test_block_error_bin_tx_0_2[31:16]	RO, NR
1.2908.15:0	PMA transmit test block error bin 2 upper	test_block_error_bin_tx_0_2[47:32]	RO, NR
1.2909.15:0	PMA transmit test block error bin 3 lower	test_block_error_bin_tx_0_3[15:0]	RO, NR
1.2910.15:0	PMA transmit test block error bin 3 middle	test_block_error_bin_tx_0_3[31:16]	RO, NR
1.2911.15:0	PMA transmit test block error bin 3 upper	test_block_error_bin_tx_0_3[47:32]	RO, NR

^a RO = Read only, NR = Non Roll-over.

Table 45–212v—Lane 0 PMA receive test block error bin register definitions (for 4 of the 17 bins)

Bit(s)	Name	Description	R/W ^a
1.3308.15:0	PMA receive test block error bin 0 lower	test_block_error_bin_rx_0_0[15:0]	RO, NR
1.3309.15:0	PMA receive test block error bin 0 middle	test_block_error_bin_rx_0_0[31:16]	RO, NR
1.3310.15:0	PMA receive test block error bin 0 upper	test_block_error_bin_rx_0_0[47:32]	RO, NR
1.3311.15:0	PMA receive test block error bin 1 lower	test_block_error_bin_rx_0_1[15:0]	RO, NR
1.3312.15:0	PMA receive test block error bin 1 middle	test_block_error_bin_rx_0_1[31:16]	RO, NR
1.3313.15:0	PMA receive test block error bin 1 upper	test_block_error_bin_rx_0_1[47:32]	RO, NR
1.3314.15:0	PMA receive test block error bin 2 lower	test_block_error_bin_rx_0_2[15:0]	RO, NR
1.3315.15:0	PMA receive test block error bin 2 middle	test_block_error_bin_rx_0_2[31:16]	RO, NR
1.3316.15:0	PMA receive test block error bin 2 upper	test_block_error_bin_rx_0_2[47:32]	RO, NR
1.3317.15:0	PMA receive test block error bin 3 lower	test_block_error_bin_rx_0_3[15:0]	RO, NR

Table 45–212v—Lane 0 PMA receive test block error bin register definitions (for 4 of the 17 bins) (*continued*)

Bit(s)	Name	Description	R/W ^a
1.3318.15:0	PMA receive test block error bin 3 middle	test_block_error_bin_rx_0_3[31:16]	RO, NR
1.3319.15:0	PMA receive test block error bin 3 upper	test_block_error_bin_rx_0_3[47:32]	RO, NR

^a RO = Read only, NR = Non Roll-over.

45.2.1.273 Registers for the upper AUI component (network side of the AUI) (Registers 1.4000 through 1.7999)

Control and status registers are required for both the upper (network side of the AUI) and lower (host side of the AUI) AUI components (see Figure 176C–3 and Figure 176D–3). Registers 1.4000 through 1.7999 have identical functionality to the registers 1.0 through 1.3999 (address offset of 4000). The relevant registers from 1.0 through 1.3999 are used for control and status of the lower AUI component. The relevant registers from 1.4000 through 1.7999 are used for control and status of the upper AUI component. Register descriptions referring to the “PMD” should be interpreted as also referring to the AUI.

45.2.3 PCS registers

45.2.3.1 PCS control 1 register (Register 3.0)

Change Table 45–234 (as amended by IEEE Std 802.3cs-2022) as follows (unchanged rows not shown):

Table 45–234—PCS control 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
3.0.5:2	Speed selection	5 4 3 2 1 1 1 * 1 1 0 1 1 1 1 1 1 1 1 0 1 1 0 1 1 1 0 0 1 0 1 1 1 0 1 0 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 0 1 0 1 0 1 0 0 0 0 1 1 0 0 1 0 0 0 0 1 0 0 0 0 = Reserved = Reserved = Reserved = 1.6 Tb/s = 800 Gb/s = 10/2.5 Gb/s = 25/10 Gb/s = 400 Gb/s = 200 Gb/s = 5 Gb/s = 2.5 Gb/s = 50 Gb/s = 25 Gb/s = 100 Gb/s = 40 Gb/s = 10/1 Gb/s = 10PASS-TS/2BASE-TL = 10 Gb/s	R/W
...			

^a RO = Read only, R/W = Read/Write, SC = Self-clearing.

45.2.3.2 PCS status 1 register (Register 3.1)

Change Table 45-235 (unchanged rows not shown), and insert 45.2.3.2.6a and 45.2.3.2.6b before 45.2.3.2.7 as follows:

Table 45–235—PCS status 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
3.1.5 3	Reserved	Value always 0	RO
<u>3.1.4</u>	<u>In-progress ability</u>	<u>1 = The PCS use_in_progress variable is true</u> <u>0 = The PCS use_in_progress variable is not true</u>	<u>RO</u>
<u>3.1.3</u>	<u>In-progress indication</u>	<u>1 = link_status has the value IN_PROGRESS</u> <u>0 = link_status does not have the value IN_PROGRESS</u>	<u>RO</u>
...			

^a RO = Read only, LL = Latching low, LH = Latching high.

45.2.3.2.6a In-progress ability (3.1.4)

When read as a one, bit 3.1.4 indicates that the PCS is able to signal IN_PROGRESS in addition to OK and FAIL in the link_status parameter of the AN_LINK.indication primitive generated by the PCS. When read as a zero, bit 3.1.4 indicates that the PCS is not able to signal IN_PROGRESS.

45.2.3.2.6b In-progress indication (3.1.3)

When read as a one, bit 3.1.3 indicates the AN_LINK.indication link_status parameter has the value IN_PROGRESS. IN_PROGRESS indicates inter-sublayer link training is not complete. When read as a zero, bit 3.1.3 indicates that the link_status parameter does not have the value IN_PROGRESS.

Change 45.2.3.2.7 as follows:

45.2.3.2.7 PCS receive link status (3.1.2)

When read as a one, bit 3.1.2 indicates that the PCS receive link is up. When read as a zero, bit 3.1.2 indicates that the PCS receive link is down. When a 10/25/40/50/100/200/400/800GBASE-R, 1.6TBASE-R, 10GBASE-W, or any MultiGBASE-T mode of operation is selected for the PCS using the PCS type selection field (3.7.4:0 ~~3.7.3:0~~), this bit is a latching low version of bit 3.32.12. When a 10GBASE-X mode of operation is selected for the PCS using the PCS type selection field (3.7.4:0 ~~3.7.3:0~~), this bit is a latching low version of bit 3.24.12. The receive link status bit shall be implemented with latching low behavior.

45.2.3.4 PCS speed ability (Register 3.4)

Change Table 45–236 (as amended by IEEE Std 802.3df-2024) as follows (unchanged rows not shown):

Table 45–236—PCS speed ability register bit definitions

Bit(s)	Name	Description	R/W ^a
3.4.15:12 3.4.15:11	Reserved for future speeds	Value always 0	RO
<u>3.4.11</u>	<u>1.6T capable</u>	<u>1 = PCS is capable of operating at 1.6 Tb/s</u> <u>0 = PCS is not capable of operating at 1.6 Tb/s</u>	<u>RO</u>
...			

^a RO = Read only

Insert 45.2.3.4.12 after 45.2.3.4.11 as follows:

45.2.3.4.12 1.6T capable (3.4.11)

When read as a one, bit 3.4.11 indicates that the PCS is able to operate at a data rate of 1.6 Tb/s. When read as a zero, bit 3.4.11 indicates that the PCS is not able to operate at a data rate of 1.6 Tb/s.

45.2.3.6 PCS control 2 register (Register 3.7)

Change Table 45–237 (as amended by IEEE Std 802.3cs-2022) as follows:

Table 45–237—PCS control 2 register bit definitions

Bit(s)	Name	Description	R/W ^a
3.7.15:5	Reserved	Value always 0	RO
3.7.4:0	PCS type selection	4 3 2 1 0 1 1 x x x = reserved 1 0 1 1 x = reserved <u>1 0 1 1 1 = Select 1.6TBASE-R PCS type</u> <u>1 0 1 1 0 = Select 800GBASE-R PCS type</u> 1 0 1 0 1 = Select 10GBASE-SP PCS type 1 0 1 0 0 = Select 10/2.5GBASE-SP PCS type 1 0 0 1 1 = Select 25GBASE-PQ PCS type 1 0 0 1 0 = Select 25/10GBASE-PQ PCS type 1 0 0 0 1 = Select 25GBASE-PQ PCS type, Tx only 1 0 0 0 0 = Select 25GBASE-PQ PCS type, Rx only 0 1 1 1 1 = Select 5GBASE-R PCS type 0 1 1 1 0 = Select 2.5GBASE-X PCS type 0 1 1 0 1 = Select 400GBASE-R PCS type 0 1 1 0 0 = Select 200GBASE-R PCS type 0 1 0 1 1 = Select 5GBASE-T PCS type 0 1 0 1 0 = Select 2.5GBASE-T PCS type 0 1 0 0 1 = Select 25GBASE-T PCS type 0 1 0 0 0 = Select 50GBASE-R PCS type 0 0 1 1 1 = Select 25GBASE-R PCS type 0 0 1 1 0 = Select 40GBASE-T PCS type 0 0 1 0 1 = Select 100GBASE-R PCS type 0 0 1 0 0 = Select 40GBASE-R PCS type 0 0 0 1 1 = Select 10GBASE-T PCS type 0 0 0 1 0 = Select 10GBASE-W PCS type 0 0 0 0 1 = Select 10GBASE-X PCS type 0 0 0 0 0 = Select 10GBASE-R PCS type	R/W

^a RO = Read only, R/W = Read/Write.

45.2.3.8 PCS status 3 register (Register 3.9)

Change Table 45–239 (as amended by IEEE Std 802.3cs-2022) as follows (unchanged rows not shown):

Table 45–239—PCS status 3 register bit definitions

Bit(s)	Name	Description	R/W ^a
<u>3.9.15:12</u> 3.9.15:10	Reserved	Value always 0	RO
<u>3.9.11</u>	<u>1.6TBASE-R capable</u>	<u>1 = PCS is able to support 1.6TBASE-R PCS type</u> <u>0 = PCS is not able to support 1.6TBASE-R PCS type</u>	<u>RO</u>
<u>3.9.10</u>	<u>800GBASE-R capable</u>	<u>1 = PCS is able to support 800GBASE-R PCS type</u> <u>0 = PCS is not able to support 800GBASE-R PCS type</u>	<u>RO</u>
...			

^a RO = Read only.

Insert 45.2.3.8.a and 45.2.3.8.b before 45.2.3.8.1 as follows:

45.2.3.8.a 1.6TBASE-R capable (3.9.11)

When read as a one, bit 3.9.11 indicates that the PCS is able to support the 1.6TBASE-R PCS type. When read as a zero, bit 3.9.11 indicates that the PCS is not able to support the 1.6TBASE-R PCS type.

45.2.3.8.b 800GBASE-R capable (3.9.10)

When read as a one, bit 3.9.10 indicates that the PCS is able to support the 800GBASE-R PCS type. When read as a zero, bit 3.9.10 indicates that the PCS is not able to support the 800GBASE-R PCS type.

45.2.3.15 BASE-R and MultiGBASE-T PCS status 1 register (Register 3.32)

Change 45.2.3.15.1 as follows:

45.2.3.15.1 BASE-R and MultiGBASE-T receive link status (3.32.12)

When read as a one, bit 3.32.12 indicates that the PCS is in a fully operational state. When read as a zero, bit 3.32.12 indicates that the PCS is not fully operational. This bit is a reflection of the PCS_status variable defined in 49.2.14.1 for 5/10/25GBASE-R, in 126.3.7.1 for 2.5GBASE-T and 5GBASE-T, in 55.3.7.1 for 10GBASE-T, in 113.3.7.1 for 25GBASE-T and 40GBASE-T, in 82.3.1 for 40/50/100GBASE-R, ~~and~~ in 119.3 for 200/400GBASE-R, in 172.3 for 800GBASE-R, and in 175.8 for 1.6TBASE-R.

45.2.3.19 BASE-R PCS test-pattern control register (Register 3.42)

Change the fourth and fifth sentences of 45.2.3.19 (as amended by IEEE Std 802.3df-2024) as follows:

Scrambled idle test patterns are defined for 25/40/50/100/200/400/800GBASE-R and 1.6TBASE-R PCS only. The test-pattern methodology is described in 49.2.8, 82.2.11, 119.2.4.9, and 172.2.4.11, and 175.2.4.11.

45.2.4 PHY XS registers

45.2.4.1 PHY XS control 1 register (Register 4.0)

Change Table 45–315 as follows (unchanged rows not shown):

Table 45–315—PHY XS control 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
4.0.5:2	Speed selection	5 4 3 2 1 1 x x = Reserved 1 0 1 1 = Reserved 1 1 1 1 = Reserved 1 1 1 0 = 1.6 Tb/s 1 1 0 1 = 800 Gb/s 1 1 0 0 = Reserved 1 0 1 1 = Reserved 1 0 1 0 = 400 Gb/s 1 0 0 1 = 200 Gb/s 1 0 0 0 = Reserved 0 1 x x = Reserved 0 0 1 x = Reserved 0 0 0 1 = Reserved 0 0 0 0 = 10 Gb/s	R/W
...			

^a R/W = Read/Write, SC = Self-clearing, RO = Read only.

45.2.4.4 PHY XS speed ability (Register 4.4)

Change Table 45–317 (as amended by IEEE Std 802.3df-2024) as follows (most unchanged rows not shown):

Table 45–317—PHY XS speed ability register bit definitions

Bit(s)	Name	Description	R/W ^a
4.4.15:12 4.4.15:11	Reserved	Value always 0	RO
4.4.11	1.6T capable	1 = PHY XS is capable of operating at 1.6 Tb/s 0 = PHY XS is not capable of operating at 1.6 Tb/s	RO
4.4.10	800G capable	1 = PHY XS is capable of operating at 800 Gb/s 0 = PHY XS is not capable of operating at 800 Gb/s	RO
...			

^a RO = Read only

Insert 45.2.4.4.aa before 45.2.4.4.a (as inserted by IEEE Std 802.3df-2024) as follows:

45.2.4.4.aa 1.6T capable (4.4.11)

When read as a one, bit 4.4.11 indicates that the PHY XS is able to operate at a data rate of 1.6 Tb/s. When read as a zero, bit 4.4.11 indicates that the PHY-XS is not able to operate at a data rate of 1.6 Tb/s.

45.2.4.12 BASE-R PHY XS status 1 register (Register 4.32)

Change 45.2.4.12.1 as follows:

45.2.4.12.1 BASE-R PHY XS receive link status (4.32.12)

When read as a one, bit 4.32.12 indicates that the PHY XS is in a fully operational state. When read as a zero, bit 4.32.12 indicates that the PHY XS is not fully operational. This bit is a reflection of the PHY XS equivalent of the PCS_status variable defined in 119.3 for 200/400GBASE-R, in 172.3 for 800GBASE-R, and in 175.8 for 1.6TBASE-R.

Change 45.2.4.13 as follows:

45.2.4.13 BASE-R PHY XS test-pattern control register (Register 4.42)

The assignment of bits in the BASE-R PHY XS test-pattern control register is shown in Table 45–338. This register is only required when the 200/400/800GBASE-R or 1.6TBASE-R capability is supported. The test-

pattern methodology is described in 119.2.4.9 for 200/400GBASE-R, in 172.2.4.11 for 800GBASE-R, and in 175.2.4.11 for 1.6TBASE-R.

45.2.5 DTE XS registers

45.2.5.1 DTE XS control 1 register (Register 5.0)

Change Table 45–340 as follows (unchanged rows not shown):

Table 45–340—DTE XS control 1 register bit definitions

Bit(s)	Name	Description	R/W ^a
...			
5.0.5:2	Speed selection	5 4 3 2 1 1 x x = Reserved 1 0 1 1 = Reserved 1 1 1 1 = Reserved 1 1 1 0 = 1.6 Tb/s 1 1 0 1 = 800 Gb/s 1 1 0 0 = Reserved 1 0 1 1 = Reserved 1 0 1 0 = 400 Gb/s 1 0 0 1 = 200 Gb/s 1 0 0 0 = Reserved 0 1 x x = Reserved 0 0 1 x = Reserved 0 0 0 1 = Reserved 0 0 0 0 = 10 Gb/s	R/W
...			

^a R/W = Read/Write, SC = Self-clearing, RO = Read only.

45.2.5.4 DTE XS speed ability (Register 5.4)

Change Table 45–342 (as amended by IEEE Std 802.3df-2024) as follows (most unchanged rows not shown):

Table 45–342— DTE XS speed ability register bit definitions

Bit(s)	Name	Description	R/W ^a
5.4.15:12 5.4.15:11	Reserved	Value always 0	RO
<u>5.4.11</u>	<u>1.6T capable</u>	<u>1 = DTE XS is capable of operating at 1.6 Tb/s</u> <u>0 = DTE XS is not capable of operating at 1.6 Tb/s</u>	<u>RO</u>
5.4.10	800G capable	1 = DTE XS is capable of operating at 800 Gb/s 0 = DTE XS is not capable of operating at 800 Gb/s	RO
...			

^a RO = Read only

Insert 45.2.5.4.aa before 45.2.5.4.a (as inserted by IEEE Std 802.3df-2024) as follows:

45.2.5.4.aa 1.6T capable (5.4.11)

When read as a one, bit 5.4.11 indicates that the DTE XS is able to operate at a data rate of 1.6 Tb/s. When read as a zero, bit 5.4.11 indicates that the DTE XS is not able to operate at a data rate of 1.6 Tb/s.

45.2.5.12 BASE-R DTE XS status 1 register (Register 5.32)

Change 45.2.5.12.1 as follows:

45.2.5.12.1 BASE-R DTE XS receive link status (5.32.12)

When read as a one, bit 5.32.12 indicates that the DTE XS is in a fully operational state. When read as a zero, bit 5.32.12 indicates that the PHY XS is not fully operational. This bit is a reflection of the DTE XS equivalent of the PCS_status variable defined in 119.3 for 200/400GBASE-R, in 172.3 for 800GBASE-R, and in 175.8 for 1.6TBASE-R.

Change 45.2.5.13 as follows:

45.2.5.13 BASE-R DTE XS test-pattern control register (Register 5.42)

The assignment of bits in the BASE-R DTE XS test-pattern control register is shown in Table 45–338. This register is only required when the 200/400/800GBASE-R or 1.6TBASE-R capability is supported. The test-pattern methodology is described in 119.2.4.9 for 200/400GBASE-R, in 172.2.4.11 for 800GBASE-R, and in 175.2.4.11 for 1.6TBASE-R.

45.2.7 Auto-Negotiation registers

Change Table 45–378 as follows (most unchanged rows not shown):

Table 45–378—Auto-Negotiation MMD registers

Register address	Register name	Subclause
...		
7.50 through 7.51 7.59	Reserved	
7.52 through 7.53	Backplane Ethernet extended technology ability advertisement	45.2.7.13a
7.54 through 7.55	Backplane Ethernet link partner extended technology ability	45.2.7.13b
7.56 through 7.59	Reserved	
7.60	EEE advertisement 1	45.2.7.14
...		

45.2.7.12 Backplane Ethernet, BASE-R copper status (Register 7.48)

Change the title of 45.2.7.12.3 (as amended by IEEE Std 802.3df-2024) as follows:

45.2.7.12.3 Negotiated Port Type (7.48.1, 7.48.2, 7.48.3, 7.48.5, 7.48.6, 7.48.8, 7.48.9, 7.48.10, 7.48.11, 7.48.12, 7.48.13, 7.48.14, 7.48.15, 7.49.0, 7.49.1, 7.49.2, 7.49.3, 7.49.4, 7.49.5, 7.49.7, 7.49.8, 7.49.9, 7.49.10, 7.49.11)

45.2.7.13 Backplane Ethernet, BASE-R copper status 2 (Register 7.49)

Change Table 45–388 (as amended by IEEE Std 802.3df-2024) as follows (some unchanged rows not shown):

**Table 45–388—Backplane Ethernet, BASE-R copper status 2 register (Register 7.49)
bit definitions**

Bit(s)	Name	Description	RO ^a
<u>7.49.15:12</u> <u>7.49.15:8</u>	Reserved	Value always 0	RO
<u>7.49.11</u>	<u>1.6TBASE-KR8 or</u> <u>1.6TBASE-CR8</u>	<u>1 = PMA/PMD is negotiated to perform 1.6TBASE-KR8 or</u> <u>1.6TBASE-CR8</u> <u>0 = PMA/PMD is not negotiated to perform 1.6TBASE-KR8</u> <u>or 1.6TBASE-CR8</u>	<u>RO</u>
<u>7.49.10</u>	<u>800GBASE-KR4 or</u> <u>800GBASE-CR4</u>	<u>1 = PMA/PMD is negotiated to perform 800GBASE-KR4 or</u> <u>800GBASE-CR4</u> <u>0 = PMA/PMD is not negotiated to perform 800GBASE-KR4</u> <u>or 800GBASE-CR4</u>	<u>RO</u>
<u>7.49.9</u>	<u>400GBASE-KR2 or</u> <u>400GBASE-CR2</u>	<u>1 = PMA/PMD is negotiated to perform 400GBASE-KR2 or</u> <u>400GBASE-CR2</u> <u>0 = PMA/PMD is not negotiated to perform 400GBASE-KR2</u> <u>or 400GBASE-CR2</u>	<u>RO</u>
<u>7.49.8</u>	<u>200GBASE-KR1 or</u> <u>200GBASE-CR1</u>	<u>1 = PMA/PMD is negotiated to perform 200GBASE-KR1 or</u> <u>200GBASE-CR1</u> <u>0 = PMA/PMD is not negotiated to perform 200GBASE-KR1</u> <u>or 200GBASE-CR1</u>	<u>RO</u>
7.49.7	800GBASE-KR8 or 800GBASE-CR8	1 = PMA/PMD is negotiated to perform 800GBASE-KR8 or 800GBASE-CR8 0 = PMA/PMD is not negotiated to perform 800GBASE-KR8 or 800GBASE-CR8	RO
...			

^a RO = Read only

Insert 45.2.7.13a and 45.2.7.13b after 45.2.7.13 as follows:

45.2.7.13a Backplane Ethernet extended technology ability advertisement register (7.52 and 7.53)

Backplane Ethernet extended abilities are advertised in the 32-bit unformatted code field of next pages with message code 2 (see Table 73A–1a). The unformatted code field of transmitted message code 2 next pages are saved in the read only Backplane Ethernet extended technology ability advertisement register. Bits 7.53.15:0 map to U31:U16 and bits 7.52.15:0 map to U15:U0.

45.2.7.13b Backplane Ethernet link partner extended technology ability register (7.54 and 7.55)

Backplane Ethernet extended abilities are indicated in the 32-bit unformatted code field of next pages with message code 2 (see Table 73A–1a). The unformatted code field of received message code 2 next pages are saved in the read only Backplane Ethernet link partner extended technology ability register. Bits 7.55.15:0 map to U31:U16 and bits 7.54.15:0 map to U15:U0.

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